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United States Patent	12408929
Kind Code	B2
Date of Patent	September 09, 2025
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Systems and methods for navigating a pin guide driver

Abstract

Systems and methods for resecting a knee joint using a navigated pin guide driver system. The navigated pin guide driver system communicates with a navigation system to aid a user in placing cut block pins into the knee joint. The navigated pin guide driver system may include a handle, a reference element that electronically communicates with the navigation system, one or more pin guide tubes that may correspond to one or more cut blocks, and a distal tip that is configured to attach to the bone.

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Appl. No.: **17/657387**

Filed: **March 31, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220257263 A1	Aug. 18, 2022

Related U.S. Application Data

continuation-in-part parent-doc US 16737054 20200108 US 11864857 child-doc US 16841927
continuation-in-part parent-doc US 16587203 20190930 US 11890066 child-doc US 16737054

division parent-doc US 16841927 20200407 US 11426178 child-doc US 17657387 us-provisional-application US 62906831 20190927

Publication Classification

Int. Cl.: **A61B17/15** (20060101); **A61B34/20** (20160101)

U.S. Cl.:

CPC **A61B17/154** (20130101); **A61B34/20** (20160201); A61B2034/2055 (20160201)

Field of Classification Search

CPC: A61B (17/154); A61B (34/20); A61B (2034/2055)

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8498744	12/2012	Odermatt et al.	N/A	N/A
8500722	12/2012	Cooper	N/A	N/A

8500728	12/2012	Newton et al.	N/A	N/A
8504201	12/2012	Moll et al.	N/A	N/A
8506555	12/2012	Ruiz Morales	N/A	N/A
8506556	12/2012	Schena	N/A	N/A
8508173	12/2012	Goldberg et al.	N/A	N/A
8512318	12/2012	Tovey et al.	N/A	N/A
8515576	12/2012	Lipow et al.	N/A	N/A
8518120	12/2012	Glerum et al.	N/A	N/A
8521331	12/2012	Itkowitz	N/A	N/A
8526688	12/2012	Groszmann et al.	N/A	N/A
8526700	12/2012	Issacs	N/A	N/A
8527094	12/2012	Kumar et al.	N/A	N/A
8528440	12/2012	Morley et al.	N/A	N/A
8532741	12/2012	Heruth et al.	N/A	N/A
8535329	12/2012	Sarin et al.	N/A	N/A
8541970	12/2012	Nowlin et al.	N/A	N/A
8548563	12/2012	Simon et al.	N/A	N/A
8549732	12/2012	Burg et al.	N/A	N/A
8551114	12/2012	Ramos de la Pena	N/A	N/A
8551116	12/2012	Julian et al.	N/A	N/A
8556807	12/2012	Scott et al.	N/A	N/A
8556979	12/2012	Glerum et al.	N/A	N/A
8560118	12/2012	Green et al.	N/A	N/A
8561473	12/2012	Blumenkranz	N/A	N/A
8562594	12/2012	Cooper et al.	N/A	N/A
8571638	12/2012	Shoham	N/A	N/A
8571710	12/2012	Coste-Maniere et al.	N/A	N/A
8573465	12/2012	Shelton, IV	N/A	N/A
8574303	12/2012	Sharkey et al.	N/A	N/A
8585420	12/2012	Burbank et al.	N/A	N/A
8594841	12/2012	Zhao et al.	N/A	N/A
8597198	12/2012	Sanborn et al.	N/A	N/A
8600478	12/2012	Verard et al.	N/A	N/A
8603077	12/2012	Cooper et al.	N/A	N/A
8611985	12/2012	Lavallee et al.	N/A	N/A
8613230	12/2012	Blumenkranz et al.	N/A	N/A
8621939	12/2013	Blumenkranz et al.	N/A	N/A
8624537	12/2013	Nowlin et al.	N/A	N/A
8630389	12/2013	Kato	N/A	N/A
8634897	12/2013	Simon et al.	N/A	N/A
8634957	12/2013	Toth et al.	N/A	N/A
8638056	12/2013	Goldberg et al.	N/A	N/A
8638057	12/2013	Goldberg et al.	N/A	N/A
8639000	12/2013	Zhao et al.	N/A	N/A
8641726	12/2013	Bonutti	N/A	N/A
8644907	12/2013	Hartmann et al.	N/A	N/A
8657809	12/2013	Schoepp	N/A	N/A
8660635	12/2013	Simon et al.	N/A	N/A
8666544	12/2013	Moll et al.	N/A	N/A

8675939	12/2013	Moctezuma de la Barrera	N/A	N/A
8678647	12/2013	Gregerson et al.	N/A	N/A
8679125	12/2013	Smith et al.	N/A	N/A
8679183	12/2013	Glerum et al.	N/A	N/A
8682413	12/2013	Lloyd	N/A	N/A
8684253	12/2013	Giordano et al.	N/A	N/A
8685098	12/2013	Glerum et al.	N/A	N/A
8693730	12/2013	Umasuthan et al.	N/A	N/A
8694075	12/2013	Groszmann et al.	N/A	N/A
8696458	12/2013	Foxlin et al.	N/A	N/A
8700123	12/2013	Okamura et al.	N/A	N/A
8706086	12/2013	Glerum	N/A	N/A
8706185	12/2013	Foley et al.	N/A	N/A
8706301	12/2013	Zhao et al.	N/A	N/A
8717430	12/2013	Simon et al.	N/A	N/A
8727618	12/2013	Maschke et al.	N/A	N/A
8734432	12/2013	Tuma et al.	N/A	N/A
8738115	12/2013	Amberg et al.	N/A	N/A
8738181	12/2013	Greer et al.	N/A	N/A
8740882	12/2013	Jun et al.	N/A	N/A
8746252	12/2013	McGrogan et al.	N/A	N/A
8749189	12/2013	Nowlin et al.	N/A	N/A
8749190	12/2013	Nowlin et al.	N/A	N/A
8761930	12/2013	Nixon	N/A	N/A
8764448	12/2013	Yang et al.	N/A	N/A
8771170	12/2013	Mesallum et al.	N/A	N/A
8781186	12/2013	Clements et al.	N/A	N/A
8781630	12/2013	Banks et al.	N/A	N/A
8784385	12/2013	Boyden et al.	N/A	N/A
8786241	12/2013	Nowlin et al.	N/A	N/A
8787520	12/2013	Baba	N/A	N/A
8792704	12/2013	Isaacs	N/A	N/A
8798231	12/2013	Notohara et al.	N/A	N/A
8800838	12/2013	Shelton, IV	N/A	N/A
8808164	12/2013	Hoffman et al.	N/A	N/A
8812077	12/2013	Dempsey	N/A	N/A
8814793	12/2013	Brabrand	N/A	N/A
8816628	12/2013	Nowlin et al.	N/A	N/A
8818105	12/2013	Myronenko et al.	N/A	N/A
8820605	12/2013	Shelton, IV	N/A	N/A
8821511	12/2013	von Jako et al.	N/A	N/A
8823308	12/2013	Nowlin et al.	N/A	N/A
8827996	12/2013	Scott et al.	N/A	N/A
8828024	12/2013	Farritor et al.	N/A	N/A
8830224	12/2013	Zhao et al.	N/A	N/A
8834489	12/2013	Cooper et al.	N/A	N/A
8834490	12/2013	Bonutti	N/A	N/A
8838270	12/2013	Druke et al.	N/A	N/A
8844789	12/2013	Shelton, IV et al.	N/A	N/A

8852210	12/2013	Selover et al.	N/A	N/A
8855822	12/2013	Bartol et al.	N/A	N/A
8858598	12/2013	Seifert et al.	N/A	N/A
8860753	12/2013	Bhandarkar et al.	N/A	N/A
8864751	12/2013	Prisco et al.	N/A	N/A
8864798	12/2013	Weiman et al.	N/A	N/A
8864833	12/2013	Glerum et al.	N/A	N/A
8867703	12/2013	Shapiro et al.	N/A	N/A
8870880	12/2013	Himmelberger et al.	N/A	N/A
8876866	12/2013	Zappacosta et al.	N/A	N/A
8880223	12/2013	Raj et al.	N/A	N/A
8882803	12/2013	Iott et al.	N/A	N/A
8883210	12/2013	Truncale et al.	N/A	N/A
8888821	12/2013	Rezach et al.	N/A	N/A
8888853	12/2013	Glerum et al.	N/A	N/A
8888854	12/2013	Glerum et al.	N/A	N/A
8894652	12/2013	Seifert et al.	N/A	N/A
8894688	12/2013	Suh	N/A	N/A
8894691	12/2013	Iott et al.	N/A	N/A
8906069	12/2013	Hansell et al.	N/A	N/A
8961537	12/2014	Leung et al.	N/A	N/A
8964934	12/2014	Ein-Gal	N/A	N/A
8992580	12/2014	Bar et al.	N/A	N/A
8996169	12/2014	Lightcap et al.	N/A	N/A
9001963	12/2014	Sowards-Emmerd et al.	N/A	N/A
9002076	12/2014	Khadem et al.	N/A	N/A
9005113	12/2014	Scott et al.	N/A	N/A
9019078	12/2014	Hamelin et al.	N/A	N/A
9023060	12/2014	Cooper et al.	N/A	N/A
9044190	12/2014	Rubner et al.	N/A	N/A
9048613	12/2014	Kong et al.	N/A	N/A
9066751	12/2014	Sasso	N/A	N/A
9066755	12/2014	Jacobs et al.	N/A	N/A
9095376	12/2014	Plabky et al.	N/A	N/A
9107683	12/2014	Hourtash et al.	N/A	N/A
9125556	12/2014	Zehavi et al.	N/A	N/A
9131986	12/2014	Greer et al.	N/A	N/A
9161760	12/2014	Saurez et al.	N/A	N/A
9215968	12/2014	Schostek et al.	N/A	N/A
9271633	12/2015	Scott et al.	N/A	N/A
9271847	12/2015	Murphy	N/A	N/A
9308050	12/2015	Kostrzewski et al.	N/A	N/A
9339277	12/2015	Jansen et al.	N/A	N/A
9380984	12/2015	Li et al.	N/A	N/A
9393039	12/2015	Lechner et al.	N/A	N/A
9398886	12/2015	Gregerson et al.	N/A	N/A
9398890	12/2015	Dong et al.	N/A	N/A
9414859	12/2015	Ballard et al.	N/A	N/A
9420975	12/2015	Gutfleisch et al.	N/A	N/A

9421019	12/2015	Plaskos et al.	N/A	N/A
9492235	12/2015	Hourtash et al.	N/A	N/A
9565997	12/2016	Scott et al.	N/A	N/A
9566120	12/2016	Malackowski et al.	N/A	N/A
9592096	12/2016	Maillet et al.	N/A	N/A
9592100	12/2016	Olson et al.	N/A	N/A
9603621	12/2016	Zhou	N/A	N/A
9629687	12/2016	Bonutti	N/A	N/A
9713506	12/2016	Fanson et al.	N/A	N/A
9750465	12/2016	Engel et al.	N/A	N/A
9757203	12/2016	Hourtash et al.	N/A	N/A
9795354	12/2016	Menegaz et al.	N/A	N/A
9808318	12/2016	Bonutti	N/A	N/A
9814535	12/2016	Bar et al.	N/A	N/A
9820783	12/2016	Donner et al.	N/A	N/A
9833265	12/2016	Donner et al.	N/A	N/A
9848922	12/2016	Tohmeh et al.	N/A	N/A
9925011	12/2017	Gombert et al.	N/A	N/A
9931025	12/2017	Graetzel et al.	N/A	N/A
9962069	12/2017	Scott et al.	N/A	N/A
9968405	12/2017	Cooper et al.	N/A	N/A
9993273	12/2017	Moctezuma de la Barrera et al.	N/A	N/A
10034717	12/2017	Miller et al.	N/A	N/A
10045828	12/2017	Abbasi et al.	N/A	N/A
10154239	12/2017	Casas	N/A	N/A
10194131	12/2018	Casas	N/A	N/A
10194990	12/2018	Amanatullah et al.	N/A	N/A
10219865	12/2018	Jansen et al.	N/A	N/A
10292768	12/2018	Lang	N/A	N/A
10326975	12/2018	Casas	N/A	N/A
10398449	12/2018	Otto et al.	N/A	N/A
2001/0036302	12/2000	Miller	N/A	N/A
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2008/0082109	12/2007	Moll et al.	N/A	N/A
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2008/0108991	12/2007	von Jako	N/A	N/A
2008/0109012	12/2007	Falco et al.	N/A	N/A
2008/0144906	12/2007	Allred et al.	N/A	N/A
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2008/0214922	12/2007	Hartmann et al.	N/A	N/A
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2008/0287771	12/2007	Anderson	N/A	N/A
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2011/0224688	12/2010	Larkin et al.	N/A	N/A
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Primary Examiner: Boles; Sameh R

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a division of U.S. application Ser. No. 16/841,927, filed on Apr. 7, 2020, which is a continuation-in-part of U.S. application Ser. No. 16/737,054, filed on Jan. 8, 2020, which is a continuation-in-part of U.S. application Ser. No. 16/587,203, filed on Sep. 30, 2019. U.S. application Ser. No. 16/737,054 also claims the benefit of U.S. Provisional Application No. 62/906,831, filed on Sep. 27, 2019. The contents of each of these applications are incorporated herein by reference in their entirety for all purposes.

FIELD
(1) The present disclosure relates to medical devices and systems, and more particularly, robotic systems and related end effectors for controlling cutting of anatomical structures of a patient, and related methods and devices.
BACKGROUND
(2) Total Knee Arthroplasty (TKA) involves placing implants on resected surfaces on the distal femur and proximal tibia. The location and orientation of the resection plans defines the location and orientation of the implant which in turn impacts patient outcomes. There is a desire among surgeons to place implants using navigation techniques, which enables them to precisely plan and place the implant more in accordance with the patient's existing anatomy. The Navigated Pin Guide addresses the challenge of using a navigation system to localize cut plans used in total knee surgery.
(3) In the manual process, cuts are performed using the following general workflow: (1) cut block placement instruments are applied to bone; (2) the placement instrument is adjusted using touch points, visual and tactile feedback; (3) cut block pins are driven into bone and the placement instruments are removed; (4) cut blocks are attached to the bone over the cut block pins; and (5) cuts are performed through the cut blocks.
(4) Currently, using navigated components may add the following to the above-noted work flow: (6) register the patient to the navigation system; (7) navigated cut block placement instruments are applied and secured to bone using pins; (8) the placement instrument is adjusted using navigation feedback; (9) cut block pins are driven into bone and the placement instruments are removed; (10)

cut blocks are attached to the bone over the cut block pins; and (11) cuts are performed through the cut blocks.
(5) Current navigation technique may suffer from the disadvantage of using a bulky navigated placement instrument, which requires additional holes in bone and follows an undesirable workflow. What is needed is a design that bypasses the need for a placement instrument and instead navigates the insertion of the pins directly.
SUMMARY
(6) Some embodiments of the present disclosure are directed to a navigated pin guide driver system. The navigated pin guide driver system may include a handle, a first pin guide tube and a reference element attached to first pin guide tube. The reference element may be configured to be in electronic communication with a navigation system. The navigated pin guide driver system may include a distal tip configured to dock into cortical bone.
(7) Some embodiments of the present disclosure are directed to a method for performing a total knee arthroplasty surgery using a navigated pin guide driver system. The method may include registering a patient to a navigation system, aligning the navigated pin guide driver system to a target area of a knee of the patient using the navigation system, driving at least one cut block pin into the target area using the navigated pin guide driver system, attaching at least one cut block over the at least one cut block pin, and performing cuts to the target area corresponding to the at least one cut block. The navigated pin guide driver system of the above-noted method may include a handle, a first pin guide tube and a reference element attached to first pin guide tube. The reference element may be configured to be in electronic communication with a navigation system. The navigated pin guide driver system may include a distal tip configured to dock into cortical bone.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in a constitute a part of this application, illustrate certain non-limiting embodiments of inventive concepts. In the drawings:
(2) FIG. **1** illustrates an embodiment of a surgical system according to some embodiments of the present disclosure;
(3) FIG. **2** illustrates a surgical robot component of the surgical system of FIG. **1** according to some embodiments of the present disclosure;
(4) FIG. **3** illustrates a camera tracking system component of the surgical system of FIG. **1** according to some embodiments of the present disclosure;
(5) FIG. **4** illustrates an embodiment of a passive end effector that is connectable to a robot arm and configured according to some embodiments of the present disclosure;
(6) FIG. **5** illustrates a medical operation in which a surgical robot and a camera system are disposed around a patient;
(7) FIG. **6** illustrates an embodiment of an end effector coupler of a robot arm configured for connection to a passive end effector according to some embodiments of the present disclosure;
(8) FIG. **7** illustrates an embodiment of a cut away of the end effector coupler of FIG. **6**;
(9) FIG. **8** illustrates a block diagram of components of a surgical system according to some embodiments of the present disclosure;
(10) FIG. **9** illustrates a block diagram of a surgical system computer platform that includes a surgical planning computer which may be separate from and operationally connected to a surgical robot or at least partially incorporated therein according to some embodiments of the present disclosure;

(11) FIG. **10** illustrates an embodiment of a C-Arm imaging device that can be used in combination with the surgical robot and passive end effector in accordance with some embodiments of the present disclosure;
(12) FIG. **11** illustrates an embodiment of an O-Arm imaging device that can be used in combination with the surgical robot and passive end effector in accordance with some embodiments of the present disclosure; and
(13) FIGS. **12-19** illustrate alternative embodiments of passive end effectors which are configured in accordance with some embodiments of the present disclosure.
(14) FIG. **20** is a screenshot of a display showing the progress of bone cuts during a surgical procedure.
(15) FIG. **21** illustrates an exemplary embodiment of a direct blade guidance system consistent with the principles of the present disclosure.
(16) FIG. **22** illustrates an exemplary of a direct blade guidance system consistent with the principles of the present disclosure.
(17) FIGS. **23-24** illustrate an exemplary embodiment of a portion of a direct blade guidance system consistent with the principles of the present disclosure.
(18) FIG. **25** illustrates an exemplary embodiment of a direct blade guidance system consistent with the principles of the present disclosure.
(19) FIG. **26** illustrates an exemplary embodiment of a direct blade guidance system consistent with the principles of the present disclosure.
(20) FIG. **27** illustrates an exemplary embodiment of a direct blade guidance system consistent with the principles of the present disclosure.
(21) FIG. **28** illustrates an exemplary embodiment of a portion of a direct blade guidance system consistent with the principles of the present disclosure.
(22) FIGS. **29-30** illustrate an exemplary embodiment of a direct blade guidance system consistent with the principles of the present disclosure.
(23) FIG. **31** illustrates an exemplary embodiment of a blade adapter consistent with the principles of the present disclosure.
(24) FIG. **32** illustrates an exemplary embodiment of a direct blade guidance system consistent with the principles of the present disclosure.
(25) FIG. **33** illustrates a flowchart for a method of performing a knee surgery consistent with the principles of the present disclosure.
(26) FIGS. **34-37** illustrate a navigated pin guide driver system consistent with the principles of the present disclosure.
(27) FIG. **38** illustrates cut block pin inserted using a navigated pin guide driver system consistent with the principles of the present disclosure.
(28) FIGS. **39-40** illustrates a cut block inserted using a navigated pin guide driver system consistent with the principles of the present disclosure.
(29) FIG. **41** illustrates a navigated pin guide driver system consistent with the principles of the present disclosure.
(30) FIG. **42** illustrates a cut block inserted using a navigated pin guide driver system consistent with the principles of the present disclosure.
(31) FIG. **43** illustrates a resected knee joint using a navigated pin guide driver system consistent with the principles of the present disclosure.
(32) FIG. **44** illustrates a navigated pin guide driver system consistent with the principles of the present disclosure.
(33) FIG. **45** illustrates a cut block inserted using a navigated pin guide driver system consistent with the principles of the present disclosure.

DETAILED DESCRIPTION

(34) Inventive concepts will now be described more fully hereinafter with reference to the

accompanying drawings, in which examples of embodiments of inventive concepts are shown. Inventive concepts may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of various present inventive concepts to those skilled in the art. It should also be noted that these embodiments are not mutually exclusive. Components from one embodiment may be tacitly assumed to be present or used in another embodiment.
(35) Various embodiments disclosed herein are directed to improvements in operation of a surgical system when performing surgical interventions requiring osteotomy. Passive end effectors are disclosed that are connectable to a robot arm positioned by a surgical robot. The passive end effectors have pairs of mechanisms that constrain movement of a tool attachment mechanism to a range of movement. The tool attachment is connectable to a surgical saw for cutting, such as a sagittal saw having an oscillating saw blade. The mechanisms may be configured to constrain a cutting plane of the saw blade to be parallel to the working plane. The surgical robot can determine a pose of the target plane based on a surgical plan defining where an anatomical structure is to be cut and based on a pose of the anatomical structure, and can generate steering information based on comparison of the pose of the target plane and the pose of the surgical saw. The steering information indicates where the passive end effector needs to be moved so the cutting plane of the saw blade becomes aligned with the target plane and the saw blade becomes positioned a distance from the anatomical structure to be cut that is within the range of movement of the tool attachment mechanism of the passive end effector.
(36) These and other related embodiments can operate to improve the precision of the guidance of the saw blade compared to other robotic and manual (e.g., jigs) solutions for surgeries. The mechanisms of the passive end effector can allow the surgeon to concentrate on interpreting the direct force feedback while cutting bones using a surgical saw that is guided by the passive end effector. The mechanisms may be planar mechanism, e.g., having 1 to 3 adequately selected degrees of freedom (e.g. one translation or rotation, two rotations, three rotations, or other combinations, etc.) end joint that is configured to constrain the cutting plane to be aligned with the target plane. The surgeon may also more accurately monitor and control the speed of bone removal based on audio and/or visual notification feedback provided through the surgical robot.
(37) These embodiments can provide guidance during joint surgeries and especially knee surgery with high precision, high rigidity, sufficient workspace and direct force feedback. As will be explained in detail below, a tracking system can be used to precisely align the cutting plane with the target plane for cutting a bone. High precision cuts may be achieved by the planar mechanisms constraining the cutting plane to remaining aligned with the target plane while a surgeon moves the saw blade along the cutting plane and directly senses force feedback of the saw blade cutting bone. Moreover, these embodiments can be rapidly deployed into surgical practices through defined changes in existing accepted surgery workflows.
(38) FIG. **1** illustrates an embodiment of a surgical system **2** according to some embodiments of the present disclosure. Prior to performance of an orthopedic surgical procedure, a three-dimensional ("3D") image scan may be taken of a planned surgical area of a patient using, e.g., the C-Arm imaging device **104** of FIG. **10** or O-Arm imaging device **106** of FIG. **11**, or from another medical imaging device such as a computed tomography (CT) image or MRI. This scan can be taken pre-operatively (e.g. few weeks before procedure, most common) or intra-operatively. However, any known 3D or 2D image scan may be used in accordance with various embodiments of the surgical system **2**. The image scan is sent to a computer platform in communication with the surgical system **2**, such as the surgical system computer platform **900** of FIG. **9** which includes the surgical robot **800** (e.g., robot **2** in FIG. **1**) and a surgical planning computer **910**. A surgeon reviewing the image scan(s) on a display device of the surgical planning computer **910** (FIG. **9**) generates a surgical plan defining a target plane where an anatomical structure of the patient is to be cut. This plane is a

function of patient anatomy constraints, selected implant and its size. In some embodiments, the surgical plan defining the target plane is planned on the 3D image scan displayed on a display device.
(39) The surgical system **2** of FIG. **1** can assist surgeons during medical procedures by, for example, holding tools, aligning tools, using tools, guiding tools, and/or positioning tools for use. In some embodiments, surgical system **2** includes a surgical robot **4** and a camera tracking system **6**. Both systems may be mechanically coupled together by any various mechanisms. Suitable mechanisms can include, but are not limited to, mechanical latches, ties, clamps, or buttresses, or magnetic or magnetized surfaces. The ability to mechanically couple surgical robot **4** and camera tracking system **6** can allow for surgical system **2** to maneuver and move as a single unit, and allow surgical system **2** to have a small footprint in an area, allow easier movement through narrow passages and around turns, and allow storage within a smaller area.
(40) An orthopedic surgical procedure may begin with the surgical system **2** moving from medical storage to a medical procedure room. The surgical system **2** may be maneuvered through doorways, halls, and elevators to reach a medical procedure room. Within the room, the surgical system **2** may be physically separated into two separate and distinct systems, the surgical robot **4** and the camera tracking system **6**. Surgical robot **4** may be positioned adjacent the patient at any suitable location to properly assist medical personnel. Camera tracking system **6** may be positioned at the base of the patient, at patient shoulders or any other location suitable to track the present pose and movement of the pose of tracks portions of the surgical robot **4** and the patient. Surgical robot **4** and camera tracking system **6** may be powered by an onboard power source and/or plugged into an external wall outlet.
(41) Surgical robot **4** may be used to assist a surgeon by holding and/or using tools during a medical procedure. To properly utilize and hold tools, surgical robot **4** may rely on a plurality of motors, computers, and/or actuators to function properly. Illustrated in FIG. **1**, robot body **8** may act as the structure in which the plurality of motors, computers, and/or actuators may be secured within surgical robot **4**. Robot body **8** may also provide support for robot telescoping support arm **16**. In some embodiments, robot body **8** may be made of any suitable material. Suitable material may be, but is not limited to, metal such as titanium, aluminum, or stainless steel, carbon fiber, fiberglass, or heavy-duty plastic. The size of robot body **8** may provide a solid platform supporting attached components, and may house, conceal, and protect the plurality of motors, computers, and/or actuators that may operate attached components.
(42) Robot base **10** may act as a lower support for surgical robot **4**. In some embodiments, robot base **10** may support robot body **8** and may attach robot body **8** to a plurality of powered wheels **12**. This attachment to wheels may allow robot body **8** to move in space efficiently. Robot base **10** may run the length and width of robot body **8**. Robot base **10** may be about two inches to about 10 inches tall. Robot base **10** may be made of any suitable material. Suitable material may be, but is not limited to, metal such as titanium, aluminum, or stainless steel, carbon fiber, fiberglass, or heavy-duty plastic or resin. Robot base **10** may cover, protect, and support powered wheels **12**.
(43) In some embodiments, as illustrated in FIG. **1**, at least one powered wheel **12** may be attached to robot base **10**. Powered wheels **12** may attach to robot base **10** at any location. Each individual powered wheel **12** may rotate about a vertical axis in any direction. A motor may be disposed above, within, or adjacent to powered wheel **12**. This motor may allow for surgical system **2** to maneuver into any location and stabilize and/or level surgical system **2**. A rod, located within or adjacent to powered wheel **12**, may be pressed into a surface by the motor. The rod, not pictured, may be made of any suitable metal to lift surgical system **2**. Suitable metal may be, but is not limited to, stainless steel, aluminum, or titanium. Additionally, the rod may comprise at the contact-surface-side end a buffer, not pictured, which may prevent the rod from slipping and/or create a suitable contact surface. The material may be any suitable material to act as a buffer. Suitable material may be, but is not limited to, a plastic, neoprene, rubber, or textured metal. The rod may

lift powered wheel **10**, which may lift surgical system **2**, to any height required to level or otherwise fix the orientation of the surgical system **2** in relation to a patient. The weight of surgical system **2**, supported through small contact areas by the rod on each wheel, prevents surgical system **2** from moving during a medical procedure. This rigid positioning may prevent objects and/or people from moving surgical system **2** by accident.

(44) Moving surgical system **2** may be facilitated using robot railing **14**. Robot railing **14** provides a person with the ability to move surgical system **2** without grasping robot body **8**. As illustrated in FIG. **1**, robot railing **14** may run the length of robot body **8**, shorter than robot body **8**, and/or may run longer the length of robot body **8**. Robot railing **14** may be made of any suitable material. Suitable material may be, but is not limited to, metal such as titanium, aluminum, or stainless steel, carbon fiber, fiberglass, or heavy-duty plastic. Robot railing **14** may further provide protection to robot body **8**, preventing objects and or personnel from touching, hitting, or bumping into robot body **8**.

(45) Robot body **8** may provide support for a Selective Compliance Articulated Robot Arm, hereafter referred to as a "SCARA." A SCARA **24** may be beneficial to use within the surgical system **2** due to the repeatability and compactness of the robotic arm. The compactness of a SCARA may provide additional space within a medical procedure, which may allow medical professionals to perform medical procedures free of excess clutter and confining areas. SCARA **24** may comprise robot telescoping support **16**, robot support arm **18**, and/or robot arm **20**. Robot telescoping support **16** may be disposed along robot body **8**. As illustrated in FIG. **1**, robot telescoping support **16** may provide support for the SCARA **24** and display **34**. In some embodiments, robot telescoping support **16** may extend and contract in a vertical direction. Robot telescoping support **16** may be made of any suitable material. Suitable material may be, but is not limited to, metal such as titanium or stainless steel, carbon fiber, fiberglass, or heavy-duty plastic. The body of robot telescoping support **16** may be any width and/or height in which to support the stress and weight placed upon it.

(46) In some embodiments, medical personnel may move SCARA **24** through a command submitted by the medical personnel. The command may originate from input received on display **34** and/or a tablet. The command may come from the depression of a switch and/or the depression of a plurality of switches. Best illustrated in FIGS. **4** and **5**, an activation assembly **60** may include a switch and/or a plurality of switches. The activation assembly **60** may be operable to transmit a move command to the SCARA **24** allowing an operator to manually manipulate the SCARA **24**. When the switch, or plurality of switches, is depressed the medical personnel may have the ability to move SCARA **24** easily. Additionally, when the SCARA **24** is not receiving a command to move, the SCARA **24** may lock in place to prevent accidental movement by personnel and/or other objects. By locking in place, the SCARA **24** provides a solid platform upon which a passive end effector **1100** and connected surgical saw **1140**, shown in FIGS. **4** and **5**, are ready for use in a medical operation.

(47) Robot support arm **18** may be disposed on robot telescoping support **16** by various mechanisms. In some embodiments, best seen in FIGS. **1** and **2**, robot support arm **18** rotates in any direction in regard to robot telescoping support **16**. Robot support arm **18** may rotate three hundred and sixty degrees around robot telescoping support **16**. Robot arm **20** may connect to robot support arm **18** at any suitable location. Robot arm **20** may attach to robot support arm **16** by various mechanisms. Suitable mechanisms may be, but is not limited to, nuts and bolts, ball and socket fitting, press fitting, weld, adhesion, screws, rivets, clamps, latches, and/or any combination thereof. Robot arm **20** may rotate in any direction in regards to robot support arm **18**, in embodiments, robot arm **20** may rotate three hundred and sixty degrees in regards to robot support arm **18**. This free rotation may allow an operator to position robot arm **20** as planned.

(48) The passive end effector **1100** in FIGS. **4** and **5** may attach to robot arm **20** in any suitable location. As will be explained in further detail below, the passive end effector **1100** includes a base,

a first mechanism, and a second mechanism. The base is configured to attach to an end effector coupler **22** of the robot arm **20** positioned by the surgical robot **4**. Various mechanisms by which the base can attach to the end effector coupler **22** can include, but are not limited to, latch, clamp, nuts and bolts, ball and socket fitting, press fitting, weld, adhesion, screws, rivets, and/or any combination thereof. The first mechanism extends between a rotatable connection to the base and a rotatable connection to a tool attachment mechanism. The second mechanism extends between a rotatable connection to the base and a rotatable connection to the tool attachment mechanism. The first and second mechanisms pivot about the rotatable connections, and may be configured to constrain movement of the tool attachment mechanism to a range of movement within a working plane. The rotatable connections may be pivot joints allowing 1 degree-of-freedom (DOF) motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions. The tool attachment mechanism is configured to connect to a surgical saw **1140** having a saw blade or saw blade directly. The surgical saw **1140** may be configured to oscillate the saw blade for cutting. The first and second mechanisms may be configured to constrain a cutting plane of the saw blade to be parallel to the working plane. Pivot joints may be preferably used for connecting the planar mechanisms when the passive end effector is to be configured to constrain motion of the saw blade to the cutting plane.

(49) The tool attachment mechanism may connect to the surgical saw **1140** or saw blade through various mechanisms that can include, but are not limited to, a screw, nut and bolt, clamp, latch, tie, press fit, or magnet. In some embodiments, a dynamic reference array **52** is attached to the passive end effector **1100**, e.g., to the tool attachment mechanism, and/or is attached to the surgical saw **1140**. Dynamic reference arrays, also referred to as "DRAs" herein, are rigid bodies which may be disposed on a patient, the surgical robot, the passive end effector, and/or the surgical saw in a navigated surgical procedure. The camera tracking system **6** or other 3D localization system is configured to track in real-time the pose (e.g., positions and rotational orientations) of tracking markers of the DRA. The tracking markers may include the illustrated arrangement of balls or other optical markers. This tracking of 3D coordinates of tracking markers can allow the surgical system **2** to determine the pose of the DRA **52** in any space in relation to the target anatomical structure of the patient **50** in FIG. **5**.

(50) As illustrated in FIG. **1**, a light indicator **28** may be positioned on top of the SCARA **24**. Light indicator **28** may illuminate as any type of light to indicate "conditions" in which surgical system **2** is currently operating. For example, the illumination of green may indicate that all systems are normal. Illuminating red may indicate that surgical system **2** is not operating normally. A pulsating light may mean surgical system **2** is performing a function. Combinations of light and pulsation may create a nearly limitless amount of combinations in which to communicate the current operating conditions, states, or other operational indications. In some embodiments, the light may be produced by LED bulbs, which may form a ring around light indicator **28**. Light indicator **28** may comprise a fully permeable material that may let light shine through the entirety of light indicator **28**.

(51) Light indicator **28** may be attached to lower display support **30**. Lower display support **30**, as illustrated in FIG. **2** may allow an operator to maneuver display **34** to any suitable location. Lower display support **30** may attach to light indicator **28** by any suitable mechanism. In embodiments, lower display support **30** may rotate about light indicator **28**. In embodiments, lower display support **30** may attach rigidly to light indicator **28**. Light indicator **28** may then rotate three hundred and sixty degrees about robot support arm **18**. Lower display support **30** may be of any suitable length, a suitable length may be about eight inches to about thirty four inches. Lower display support **30** may act as a base for upper display support **32**.

(52) Upper display support **32** may attach to lower display support **30** by any suitable mechanism. Upper display support **32** may be of any suitable length, a suitable length may be about eight inches to about thirty four inches. In embodiments, as illustrated in FIG. **1**, upper display support

32 may allow display **34** to rotate three hundred and sixty degrees in relation to upper display support **32**. Likewise, upper display support **32** may rotate three hundred and sixty degrees in relation to lower display support **30**.

(53) Display **34** may be any device which may be supported by upper display support **32**. In embodiments, as illustrated in FIG. **2**, display **34** may produce color and/or black and white images. The width of display **34** may be about eight inches to about thirty inches wide. The height of display **34** may be about six inches to about twenty two inches tall. The depth of display **34** may be about one-half inch to about four inches.

(54) In embodiments, a tablet may be used in conjunction with display **34** and/or without display **34**. In embodiments, the table may be disposed on upper display support **32**, in place of display **34**, and may be removable from upper display support **32** during a medical operation. In addition the tablet may communicate with display **34**. The tablet may be able to connect to surgical robot **4** by any suitable wireless and/or wired connection. In some embodiments, the tablet may be able to program and/or control surgical system **2** during a medical operation. When controlling surgical system **2** with the tablet, all input and output commands may be duplicated on display **34**. The use of a tablet may allow an operator to manipulate surgical robot **4** without having to move around patient **50** and/or to surgical robot **4**.

(55) As illustrated in FIG. **5**, camera tracking system **6** works in conjunction with surgical robot **4** through wired or wireless communication networks. Referring to FIGS. **1** and **5**, camera tracking system **6** can include some similar components to the surgical robot **4**. For example, camera body **36** may provide the functionality found in robot body **8**. Robot body **8** may provide the structure upon which camera **46** is mounted. The structure within robot body **8** may also provide support for the electronics, communication devices, and power supplies used to operate camera tracking system **6**. Camera body **36** may be made of the same material as robot body **8**. Camera tracking system **6** may communicate directly to the tablet and/or display **34** by a wireless and/or wired network to enable the tablet and/or display **34** to control the functions of camera tracking system **6**.

(56) Camera body **36** is supported by camera base **38**. Camera base **38** may function as robot base **10**. In the embodiment of FIG. **1**, camera base **38** may be wider than robot base **10**. The width of camera base **38** may allow for camera tracking system **6** to connect with surgical robot **4**. As illustrated in FIG. **1**, the width of camera base **38** may be large enough to fit outside robot base **10**. When camera tracking system **6** and surgical robot **4** are connected, the additional width of camera base **38** may allow surgical system **2** additional maneuverability and support for surgical system **2**.

(57) As with robot base **10**, a plurality of powered wheels **12** may attach to camera base **38**. Powered wheel **12** may allow camera tracking system **6** to stabilize and level or set fixed orientation in regards to patient **50**, similar to the operation of robot base **10** and powered wheels **12**. This stabilization may prevent camera tracking system **6** from moving during a medical procedure and may keep camera **46** from losing track of one or more DRAs **52** connected to an anatomical structure **54** and/or tool **58** within a designated area **56** as shown in FIG. **5**. This stability and maintenance of tracking enhances the ability of surgical robot **4** to operate effectively with camera tracking system **6**. Additionally, the wide camera base **38** may provide additional support to camera tracking system **6**. Specifically, a wide camera base **38** may prevent camera tracking system **6** from tipping over when camera **46** is disposed over a patient, as illustrated in FIG. **5**. Without the wide camera base **38**, the outstretched camera **46** may unbalance camera tracking system **6**, which may result in camera tracking system **6** falling over.

(58) Camera telescoping support **40** may support camera **46**. In embodiments, telescoping support **40** may move camera **46** higher or lower in the vertical direction. Telescoping support **40** may be made of any suitable material in which to support camera **46**. Suitable material may be, but is not limited to, metal such as titanium, aluminum, or stainless steel, carbon fiber, fiberglass, or heavy-duty plastic. Camera handle **48** may be attached to camera telescoping support **40** at any suitable location. Cameral handle **48** may be any suitable handle configuration. A suitable configuration

may be, but is not limited to, a bar, circular, triangular, square, and/or any combination thereof. As illustrated in FIG. **1**, camera handle **48** may be triangular, allowing an operator to move camera tracking system **6** into a planned position before a medical operation. In embodiments, camera handle **48** may be used to lower and raise camera telescoping support **40**. Camera handle **48** may perform the raising and lowering of camera telescoping support **40** through the depression of a button, switch, lever, and/or any combination thereof.

(59) Lower camera support arm **42** may attach to camera telescoping support **40** at any suitable location, in embodiments, as illustrated in FIG. **1**, lower camera support arm **42** may rotate three hundred and sixty degrees around telescoping support **40**. This free rotation may allow an operator to position camera **46** in any suitable location. Lower camera support arm **42** may be made of any suitable material in which to support camera **46**. Suitable material may be, but is not limited to, metal such as titanium, aluminum, or stainless steel, carbon fiber, fiberglass, or heavy-duty plastic. Cross-section of lower camera support arm **42** may be any suitable shape. Suitable cross-sectional shape may be, but is not limited to, circle, square, rectangle, hexagon, octagon, or i-beam. The cross-sectional length and width may be about one to ten inches. Length of the lower camera support arm may be about four inches to about thirty-six inches. Lower camera support arm **42** may connect to telescoping support **40** by any suitable mechanism. Suitable mechanism may be, but is not limited to, nuts and bolts, ball and socket fitting, press fitting, weld, adhesion, screws, rivets, clamps, latches, and/or any combination thereof. Lower camera support arm **42** may be used to provide support for camera **46**. Camera **46** may be attached to lower camera support arm **42** by any suitable mechanism. Suitable mechanism may be, but is not limited to, nuts and bolts, ball and socket fitting, press fitting, weld, adhesion, screws, rivets, and/or any combination thereof. Camera **46** may pivot in any direction at the attachment area between camera **46** and lower camera support arm **42**. In embodiments a curved rail **44** may be disposed on lower camera support arm **42**.

(60) Curved rail **44** may be disposed at any suitable location on lower camera support arm **42**. As illustrated in FIG. **3**, curved rail **44** may attach to lower camera support arm **42** by any suitable mechanism. Suitable mechanism may be, but are not limited to nuts and bolts, ball and socket fitting, press fitting, weld, adhesion, screws, rivets, clamps, latches, and/or any combination thereof. Curved rail **44** may be of any suitable shape, a suitable shape may be a crescent, circular, oval, elliptical, and/or any combination thereof. In embodiments, curved rail **44** may be any appropriate length. An appropriate length may be about one foot to about six feet. Camera **46** may be moveably disposed along curved rail **44**. Camera **46** may attach to curved rail **44** by any suitable mechanism. Suitable mechanism may be, but are not limited to rollers, brackets, braces, motors, and/or any combination thereof. Motors and rollers, not illustrated, may be used to move camera **46** along curved rail **44**. As illustrated in FIG. **3**, during a medical procedure, if an object prevents camera **46** from viewing one or more DRAs **52**, the motors may move camera **46** along curved rail **44** using rollers. This motorized movement may allow camera **46** to move to a new position that is no longer obstructed by the object without moving camera tracking system **6**. While camera **46** is obstructed from viewing DRAs **52**, camera tracking system **6** may send a stop signal to surgical robot **4**, display **34**, and/or a tablet. The stop signal may prevent SCARA **24** from moving until camera **46** has reacquired DRAs **52**. This stoppage may prevent SCARA **24** and/or end effector coupler **22** from moving and/or using medical tools without being tracked by surgical system **2**.

(61) End effector coupler **22**, as illustrated in FIG. **6**, is configured to connect various types of passive end effectors to surgical robot **4**. End effector coupler **22** can include a saddle joint **62**, an activation assembly **60**, a load cell **64** (FIG. **7**), and a connector **66**. Saddle joint **62** may attach end effector coupler **22** to SCARA **24**. Saddle joint **62** may be made of any suitable material. Suitable material may be, but is not limited to metal such as titanium, aluminum, or stainless steel, carbon fiber, fiberglass, or heavy-duty plastic. Saddle joint **62** may be made of a single piece of metal which may provide end effector with additional strength and durability. The saddle joint **62** may attach to SCARA **24** by an attachment point **68**. There may be a plurality of attachment points **68**

disposed about saddle joint **62**. Attachment points **68** may be sunk, flush, and/or disposed upon saddle joint **62**. In some examples, screws, nuts and bolts, and/or any combination thereof may pass through attachment point **68** and secure saddle joint **62** to SCARA **24**. The nuts and bolts may connect saddle joint **62** to a motor, not illustrated, within SCARA **24**. The motor may move saddle joint **62** in any direction. The motor may further prevent saddle joint **62** from moving from accidental bumps and/or accidental touches by actively servoing at the current location or passively by applying spring actuated brakes.
(62) The end effector coupler **22** can include a load cell **64** interposed between the saddle join **62** and a connected passive end effector. Load cell **64**, as illustrated in FIG. **7** may attach to saddle joint **62** by any suitable mechanism. Suitable mechanism may be, but is not limited to, screws, nuts and bolts, threading, press fitting, and/or any combination thereof.
(63) FIG. **8** illustrates a block diagram of components of a surgical system **800** according to some embodiments of the present disclosure. Referring to FIGS. **7** and **8**, load cell **64** may be any suitable instrument used to detect and measure forces. In some examples, load cell **64** may be a six axis load cell, a three-axis load cell or a uniaxial load cell. Load cell **64** may be used to track the force applied to end effector coupler **22**. In some embodiments the load cell **64** may communicate with a plurality of motors **850**, **851**, **852**, **853**, and/or **854**. As load cell **64** senses force, information as to the amount of force applied may be distributed from a switch array and/or a plurality of switch arrays to a controller **846**. Controller **846** may take the force information from load cell **64** and process it with a switch algorithm. The switch algorithm is used by the controller **846** to control a motor driver **842**. The motor driver **842** controls operation of one or more of the motors. Motor driver **842** may direct a specific motor to produce, for example, an equal amount of force measured by load cell **64** through the motor. In some embodiments, the force produced may come from a plurality of motors, e.g., **850-854**, as directed by controller **846**. Additionally, motor driver **842** may receive input from controller **846**. Controller **846** may receive information from load cell **64** as to the direction of force sensed by load cell **64**. Controller **846** may process this information using a motion controller algorithm. The algorithm may be used to provide information to specific motor drivers **842**. To replicate the direction of force, controller **846** may activate and/or deactivate certain motor drivers **842**. Controller **846** may control one or more motors, e.g. one or more of **850-854**, to induce motion of passive end effector **1100** in the direction of force sensed by load cell **64**. This force-controlled motion may allow an operator to move SCARA **24** and passive end effector **1100** effortlessly and/or with very little resistance. Movement of passive end effector **1100** can be performed to position passive end effector **1100** in any suitable pose (i.e., location and angular orientation relative to defined three-dimensional (3D) orthogonal reference axes) for use by medical personnel.
(64) Connector **66** is configured to be connectable to the base of the passive end effector **1100** and is connected to load cell **64**. Connector **66** can include attachment points **68**, a sensory button **70**, tool guides **72**, and/or tool connections **74**. Best illustrated in FIGS. **6** and **8**, there may be a plurality of attachment points **68**. Attachment points **68** may connect connector **66** to load cell **64**. Attachment points **68** may be sunk, flush, and/or disposed upon connector **66**. Attachment points **68** and **76** can be used to attach connector **66** to load cell **64** and/or to passive end effector **1100**. In some examples, Attachment points **68** and **76** may include screws, nuts and bolts, press fittings, magnetic attachments, and/or any combination thereof.
(65) As illustrated in FIG. **6**, a sensory button **70** may be disposed about center of connector **66**. Sensory button **70** may be depressed when a passive end effector **1100** is connected to SCARA **24**. Depression of sensory button **70** may alert surgical robot **4**, and in turn medical personnel, that a passive end effector **1100** has been attached to SCARA **24**. As illustrated in FIG. **6**, guides **72** may be used to facilitate proper attachment of passive end effector **1100** to SCARA **24**. Guides **72** may be sunk, flush, and/or disposed upon connector **66**. In some examples there may be a plurality of guides **72** and may have any suitable patterns and may be oriented in any suitable direction. Guides

72 may be any suitable shape to facilitate attachment of passive end effector **1100** to SCARA **24**. A suitable shape may be, but is not limited to, circular, oval, square, polyhedral, and/or any combination thereof. Additionally, guides **72** may be cut with a bevel, straight, and/or any combination thereof.

(66) Connector **66** may have attachment points **74**. As illustrated in FIG. **6**, attachment points **74** may form a ledge and/or a plurality of ledges. Attachment points **74** may provide connector **66** a surface upon which passive end effector **1100** may clamp. In some embodiments, attachment points **74** are disposed about any surface of connector **66** and oriented in any suitable manner in relation to connector **66**.

(67) Activation assembly **60**, best illustrated in FIGS. **6** and **7**, may encircle connector **66**. In some embodiments, activation assembly **60** may take the form of a bracelet that wraps around connector **66**. In some embodiments, activation assembly **60**, may be located in any suitable area within surgical system **2**. In some examples, activation assembly **60** may be located on any part of SCARA **24**, any part of end effector coupler **22**, may be worn by medical personnel (and communicate wirelessly), and/or any combination thereof. Activation assembly **60** may be made of any suitable material. Suitable material may be, but is not limited to neoprene, plastic, rubber, gel, carbon fiber, fabric, and/or any combination thereof. Activation assembly **60** may comprise of a primary button **78** and a secondary button **80**. Primary button **78** and secondary button **80** may encircle the entirety of connector **66**.

(68) Primary button **78** may be a single ridge, as illustrated in FIG. **6**, which may encircle connector **66**. In some examples, primary button **78** may be disposed upon activation assembly **60** along the end farthest away from saddle joint **62**. Primary button **78** may be disposed upon primary activation switch **82**, best illustrated on FIG. **7**. Primary activation switch **82** may be disposed between connector **66** and activation assembly **60**. In some examples, there may be a plurality of primary activation switches **82**, which may be disposed adjacent and beneath primary button **78** along the entire length of primary button **78**. Depressing primary button **78** upon primary activation switch **82** may allow an operator to move SCARA **24** and end effector coupler **22**. As discussed above, once set in place, SCARA **24** and end effector coupler **22** may not move until an operator programs surgical robot **4** to move SCARA **24** and end effector coupler **22**, or is moved using primary button **78** and primary activation switch **82**. In some examples, it may require the depression of at least two non-adjacent primary activation switches **82** before SCARA **24** and end effector coupler **22** will respond to operator commands. Depression of at least two primary activation switches **82** may prevent the accidental movement of SCARA **24** and end effector coupler **22** during a medical procedure.

(69) Activated by primary button **78** and primary activation switch **82**, load cell **64** may measure the force magnitude and/or direction exerted upon end effector coupler **22** by an operator, i.e. medical personnel. This information may be transferred to motors within SCARA **24** that may be used to move SCARA **24** and end effector coupler **22**. Information as to the magnitude and direction of force measured by load cell **64** may cause the motors to move SCARA **24** and end effector coupler **22** in the same direction as sensed by load cell **64**. This force-controlled movement may allow the operator to move SCARA **24** and end effector coupler **22** easily and without large amounts of exertion due to the motors moving SCARA **24** and end effector coupler **22** at the same time the operator is moving SCARA **24** and end effector coupler **22**.

(70) Secondary button **80**, as illustrated in FIG. **6**, may be disposed upon the end of activation assembly **60** closest to saddle joint **62**. In some examples secondary button **80** may comprise a plurality of ridges. The plurality of ridges may be disposed adjacent to each other and may encircle connector **66**. Additionally, secondary button **80** may be disposed upon secondary activation switch **84**. Secondary activation switch **84**, as illustrated in FIG. **7**, may be disposed between secondary button **80** and connector **66**. In some examples, secondary button **80** may be used by an operator as a “selection” device. During a medical operation, surgical robot **4** may notify medical personnel to

certain conditions by display **34** and/or light indicator **28**. Medical personnel may be prompted by surgical robot **4** to select a function, mode, and/or assess the condition of surgical system **2**. Depressing secondary button **80** upon secondary activation switch **84** a single time may activate certain functions, modes, and/or acknowledge information communicated to medical personnel through display **34** and/or light indicator **28**. Additionally, depressing secondary button **80** upon secondary activation switch **84** multiple times in rapid succession may activate additional functions, modes, and/or select information communicated to medical personnel through display **34** and/or light indicator **28**. In some examples, at least two non-adjacent secondary activation switches **84** may be depressed before secondary button **80** may function properly. This requirement may prevent unintended use of secondary button **80** from accidental bumping by medical personnel upon activation assembly **60**. Primary button **78** and secondary button **80** may use software architecture **86** to communicate commands of medical personnel to surgical system **2**.

(71) FIG. **8** illustrates a block diagram of components of a surgical system **800** configured according to some embodiments of the present disclosure, and which may correspond to the surgical system **2** above. Surgical system **800** includes platform subsystem **802**, computer subsystem **820**, motion control subsystem **840**, and tracking subsystem **830**. Platform subsystem **802** includes battery **806**, power distribution module **804**, connector panel **808**, and charging station **810**. Computer subsystem **820** includes computer **822**, display **824**, and speaker **826**. Motion control subsystem **840** includes driver circuit **842**, motors **850**, **851**, **852**, **853**, **854**, stabilizers **855**, **856**, **857**, **858**, end effector connector **844**, and controller **846**. Tracking subsystem **830** includes position sensor **832** and camera converter **834**. Surgical system **800** may also include a removable foot pedal **880** and removable tablet computer **890**.

(72) Input power is supplied to surgical system **800** via a power source which may be provided to power distribution module **804**. Power distribution module **804** receives input power and is configured to generate different power supply voltages that are provided to other modules, components, and subsystems of surgical system **800**. Power distribution module **804** may be configured to provide different voltage supplies to connector panel **808**, which may be provided to other components such as computer **822**, display **824**, speaker **826**, driver **842** to, for example, power motors **850-854** and end effector coupler **844**, and provided to camera converter **834** and other components for surgical system **800**. Power distribution module **804** may also be connected to battery **806**, which serves as temporary power source in the event that power distribution module **804** does not receive power from an input power. At other times, power distribution module **804** may serve to charge battery **806**.

(73) Connector panel **808** may serve to connect different devices and components to surgical system **800** and/or associated components and modules. Connector panel **808** may contain one or more ports that receive lines or connections from different components. For example, connector panel **808** may have a ground terminal port that may ground surgical system **800** to other equipment, a port to connect foot pedal **880**, a port to connect to tracking subsystem **830**, which may include position sensor **832**, camera converter **834**, and marker tracking cameras **870**. Connector panel **808** may also include other ports to allow USB, Ethernet, HDMI communications to other components, such as computer **822**.

(74) Control panel **816** may provide various buttons or indicators that control operation of surgical system **800** and/or provide information from surgical system **800** for observation by an operator. For example, control panel **816** may include buttons to power on or off surgical system **800**, lift or lower vertical column **16**, and lift or lower stabilizers **855-858** that may be designed to engage casters **12** to lock surgical system **800** from physically moving. Other buttons may stop surgical system **800** in the event of an emergency, which may remove all motor power and apply mechanical brakes to stop all motion from occurring. Control panel **816** may also have indicators notifying the operator of certain system conditions such as a line power indicator or status of charge for battery **806**.

(75) Computer **822** of computer subsystem **820** includes an operating system and software to operate assigned functions of surgical system **800**. Computer **822** may receive and process information from other components (for example, tracking subsystem **830**, platform subsystem **802**, and/or motion control subsystem **840**) in order to display information to the operator. Further, computer subsystem **820** may provide output through the speaker **826** for the operator. The speaker may be part of the surgical robot, part of a head-mounted display component, or within another component of the surgical system **2**. The display **824** may correspond to the display **34** shown in FIGS. **1** and **2**, or may be a head-mounted display which projects images onto a see-through display screen which forms an augmented reality (AR) image that is overlaid on real-world objects viewable through the see-through display screen.
(76) Tracking subsystem **830** may include position sensor **832** and camera converter **834**. Tracking subsystem **830** may correspond to the camera tracking system **6** of FIG. **3**. The marker tracking cameras **870** operate with the position sensor **832** to determine the pose of DRAs **52**. This tracking may be conducted in a manner consistent with the present disclosure including the use of infrared or visible light technology that tracks the location of active or passive elements of DRAs **52**, such as LEDs or reflective markers, respectively. The location, orientation, and position of structures having these types of markers, such as DRAs **52**, is provided to computer **822** and which may be shown to an operator on display **824**. For example, as shown in FIGS. **4** and **5**, a surgical saw **1240** having a DRA **52** or which is connected to an end effector coupler **22** having a DRA **52** tracked in this manner (which may be referred to as a navigational space) may be shown to an operator in relation to a three dimensional image of a patient's anatomical structure.
(77) Motion control subsystem **840** may be configured to physically move vertical column **16**, upper arm **18**, lower arm **20**, or rotate end effector coupler **22**. The physical movement may be conducted through the use of one or more motors **850-854**. For example, motor **850** may be configured to vertically lift or lower vertical column **16**. Motor **851** may be configured to laterally move upper arm **18** around a point of engagement with vertical column **16** as shown in FIG. **2**. Motor **852** may be configured to laterally move lower arm **20** around a point of engagement with upper arm **18** as shown in FIG. **2**. Motors **853** and **854** may be configured to move end effector coupler **22** to provide translational movement and rotation along in about three-dimensional axes. The surgical planning computer **910** shown in FIG. **9** can provide control input to the controller **846** that guides movement of the end effector coupler **22** to position a passive end effector, which is connected thereto, with a planned pose (i.e., location and angular orientation relative to defined 3D orthogonal reference axes) relative to an anatomical structure that is to be cut during a surgical procedure. Motion control subsystem **840** may be configured to measure position of the passive end effector structure using integrated position sensors (e.g. encoders). In one of the embodiments, position sensors are directly connected to at least one joint of the passive end effector structure, but may also be positioned in another location in the structure and remotely measure the joint position by interconnection of a timing belt, a wire, or any other synchronous transmission interconnection.
(78) FIG. **9** illustrates a block diagram of a surgical system computer platform **900** that includes a surgical planning computer **910** which may be separate from and operationally connected to a surgical robot **800** or at least partially incorporated therein according to some embodiments of the present disclosure. Alternatively, at least a portion of operations disclosed herein for the surgical planning computer **910** may be performed by components of the surgical robot **800** such as by the computer subsystem **820**.
(79) Referring to FIG. **9**, the surgical planning computer **910** includes a display **912**, at least one processor circuit **914** (also referred to as a processor for brevity), at least one memory circuit **916** (also referred to as a memory for brevity) containing computer readable program code **918**, and at least one network interface **920** (also referred to as a network interface for brevity). The network interface **920** can be configured to connect to a C-Arm imaging device **104** in FIG. **10**, an O-Arm imaging device **106** in FIG. **11**, another medical imaging device, an image database **950** of medical

images, components of the surgical robot **800**, and/or other electronic equipment.
(80) When the surgical planning computer **910** is at least partially integrated within the surgical robot **800**, the display **912** may correspond to the display **34** of FIG. **2** and/or the tablet **890** of FIG. **8** and/or a head-mounted display, the network interface **920** may correspond to the platform network interface **812** of FIG. **8**, and the processor **914** may correspond to the computer **822** of FIG. **8**.
(81) The processor **914** may include one or more data processing circuits, such as a general purpose and/or special purpose processor, e.g., microprocessor and/or digital signal processor. The processor **914** is configured to execute the computer readable program code **918** in the memory **916** to perform operations, which may include some or all of the operations described herein as being performed by a surgical planning computer.
(82) The processor **914** can operate to display on the display device **912** an image of a bone that is received from one of the imaging devices **104** and **106** and/or from the image database **950** through the network interface **920**. The processor **914** receives an operator's definition of where an anatomical structure, i.e. one or more bones, shown in one or more images is to be cut, such as by an operator touch selecting locations on the display **912** for planned surgical cuts or using a mouse-based cursor to define locations for planned surgical cuts.
(83) The surgical planning computer **910** enables anatomy measurement, useful for knee surgery, like measurement of various angles determining center of hip, center of angles, natural landmarks (e.g. transepicondylar line, Whitesides line, posterior condylar line etc.), etc. Some measurements can be automatic while some others involve human input or assistance. This surgical planning computer **910** allows an operator to choose the correct implant for a patient, including choice of size and alignment. The surgical planning computer **910** enables automatic or semi-automatic (involving human input) segmentation (image processing) for CT images or other medical images. The surgical plan for a patient may be stored in a cloud-based server for retrieval by the surgical robot **800**. During the surgery, the surgeon will choose which cut to make (e.g. posterior femur, proximal tibia etc.) using a computer screen (e.g. touchscreen) or augmented reality interaction via, e.g., a head-mounted display. The surgical robot **4** may automatically move the surgical saw blade to a planned position so that a target plane of planned cut is optimally placed within a workspace of the passive end effector interconnecting the surgical saw blade and the robot arm **20**. Command enabling movement can be given by user using various modalities, e.g. foot pedal.
(84) In some embodiments, the surgical system computer platform **900** can use two DRAs to tracking patient anatomy position: one on patient tibia and one on patient femur. The platform **900** may use standard navigated instruments for the registration and checks (e.g., a pointer similar to the one used in Globus ExcelsiusGPS system for spine surgery). Tracking markers allowing for detection of DRAs movement in reference to tracked anatomy can be used as well.
(85) An important difficulty in knee surgery is how to plan the position of the implant in the knee and many surgeons struggle with to do it on a computer screen which is a 2D representation of 3D anatomy. The platform **900** could address this problem by using an augmented reality (AR) head-mounted display to generate an implant overlay around the actual patient knee. For example, the surgeon can be operationally displayed a virtual handle to grab and move the implant to a desired pose and adjust planned implant placement. Afterward, during surgery, the platform **900** could render the navigation through the AR head-mounted display to show surgeon what is not directly visible. Also, the progress of bone removal, e.g., depth or cut, can be displayed in real-time. Other features that may be displayed through AR can include, without limitation, gap or ligament balance along a range of joint motion, contact line on the implant along the range of joint motion, ligament tension and/or laxity through color or other graphical overlays, etc.
(86) The surgical planning computer **910**, in some embodiments, can allow planning for use of standard implants, e.g., posterior stabilized implants and cruciate retaining implants, cemented and cementless implants, revision systems for surgeries related to, for example, total or partial knee

and/or hip replacement and/or trauma.

(87) The processor **912** may graphically illustrate on the display **912** one or more cutting planes intersecting the displayed anatomical structure at the locations selected by the operator for cutting the anatomical structure. The processor **912** also determines one or more sets of angular orientations and locations where the end effector coupler **22** must be positioned so a cutting plane of the surgical saw blade will be aligned with a target plane to perform the operator defined cuts, and stores the sets of angular orientations and locations as data in a surgical plan data structure. The processor **912** uses the known range of movement of the tool attachment mechanism of the passive end effector to determine where the end effector coupler **22** attached to the robot arm **20** needs to be positioned.

(88) The computer subsystem **820** of the surgical robot **800** receives data from the surgical plan data structure and receives information from the camera tracking system **6** indicating a present pose of an anatomical structure that is to be cut and indicating a present pose of the passive end effector and/or surgical saw tracked through DRAs. The computer subsystem **820** determines a pose of the target plane based on the surgical plan defining where the anatomical structure is to be cut and based on the pose of the anatomical structure. The computer subsystem **820** generates steering information based on comparison of the pose of the target plane and the pose of the surgical saw. The steering information indicates where the passive end effector needs to be moved so the cutting plane of the saw blade becomes aligned with the target plane and the saw blade becomes positioned a distance from the anatomical structure to be cut that is within the range of movement of the tool attachment mechanism of the passive end effector.

(89) As explained above, a surgical robot includes a robot base, a robot arm connected to the robot base, and at least one motor operatively connected to move the robot arm relative to the robot base. The surgical robot also includes at least one controller, e.g. the computer subsystem **820** and the motion control subsystem **840**, connected to the at least one motor and configured to perform operations.

(90) As will be explained in further detail below with regard to FIGS. **12-19**, a passive end effector includes a base configured to attach to an activation assembly of the robot arm, a first mechanism, and a second mechanism. The first mechanism extends between a rotatable connection to the base and a rotatable connection to a tool attachment mechanism. The second mechanism extends between a rotatable connection to the base and a rotatable connection to the tool attachment mechanism. The first and second mechanisms pivot about the rotatable connections which may be configured to constrain movement of the tool attachment mechanism to a range of movement within a working plane. The rotatable connections may be pivot joints allowing 1 degree-of-freedom (DOF) motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions. The tool attachment mechanism is configured to connect to the surgical saw comprising a saw blade for cutting. The first and second mechanisms may be configured to constrain a cutting plane of the saw blade to be parallel to the working plane.

(91) In some embodiments, the operations performed by the at least one controller of the surgical robot also includes controlling movement of the at least one motor based on the steering information to reposition the passive end effector so the cutting plane of the saw blade becomes aligned with the target plane and the saw blade becomes positioned the distance from the anatomical structure to be cut that is within the range of movement of the tool attachment mechanism of the passive end effector. The steering information may be displayed to guide an operator's movement of the surgical saw and/or may be used by the at least one controller to automatically move the surgical saw.

(92) In one embodiment, the operations performed by the at least one controller of the surgical robot also includes providing the steering information to a display device for display to guide operator movement of the passive end effector so the cutting plane of the saw blade becomes aligned with the target plane and so the saw blade becomes positioned the distance from the

anatomical structure, which is to be cut, that is within the range of movement of the tool attachment mechanism of the passive end effector. The display device may correspond to the display **824** (FIG. **8**), the display **34** of FIG. **1**, and/or a head-mounted display.
(93) For example, the steering information may be displayed on a head-mounted display which projects images onto a see-through display screen which forms an augmented reality image that is overlaid on real-world objects viewable through the see-through display screen. The operations may display a graphical representation of the target plane with a pose overlaid on a bone and with a relative orientation there between corresponding to the surgical plan for how the bone is planned to be cut. The operations may alternatively or additionally display a graphical representation of the cutting plane of the saw blade so that an operator may more easily align the cutting plane with the planned target plane for cutting the bone. The operator may thereby visually observe and perform movements to align the cutting plane of the saw blade with the target plane so the saw blade becomes positioned at the planned pose relative to the bone and within a range of movement of the tool attachment mechanism of the passive end effector.
(94) An automated imaging system can be used in conjunction with the surgical planning computer **910** and/or the surgical system **2** to acquire pre-operative, intra-operative, post-operative, and/or real-time image data of a patient. Example automated imaging systems are illustrated in FIGS. **10** and **11**. In some embodiments, the automated imaging system is a C-arm **104** (FIG. **10**) imaging device or an O-arm® **106** (FIG. **11**). (O-arm® is copyrighted by Medtronic Navigation, Inc. having a place of business in Louisville, Colo., USA) It may be desirable to take x-rays of a patient from a number of different positions, without the need for frequent manual repositioning of the patient which may be required in an x-ray system. C-arm **104** x-ray diagnostic equipment may solve the problems of frequent manual repositioning and may be well known in the medical art of surgical and other interventional procedures. As illustrated in FIG. **10**, a C-arm includes an elongated C-shaped member terminating in opposing distal ends **112** of the “C” shape. C-shaped member is attached to an x-ray source **114** and an image receptor **116**. The space within C-arm **104** of the arm provides room for the physician to attend to the patient substantially free of interference from the x-ray support structure.
(95) The C-arm is mounted to enable rotational movement of the arm in two degrees of freedom, (i.e. about two perpendicular axes in a spherical motion). C-arm is slidably mounted to an x-ray support structure, which allows orbiting rotational movement of the C-arm about its center of curvature, which may permit selective orientation of x-ray source **114** and image receptor **116** vertically and/or horizontally. The C-arm may also be laterally rotatable, (i.e. in a perpendicular direction relative to the orbiting direction to enable selectively adjustable positioning of x-ray source **114** and image receptor **116** relative to both the width and length of the patient). Spherically rotational aspects of the C-arm apparatus allow physicians to take x-rays of the patient at an optimal angle as determined with respect to the particular anatomical condition being imaged.
(96) The O-arm® **106** illustrated in FIG. **11** includes a gantry housing **124** which may enclose an image capturing portion, not illustrated. The image capturing portion includes an x-ray source and/or emission portion and an x-ray receiving and/or image receiving portion, which may be disposed about one hundred and eighty degrees from each other and mounted on a rotor (not illustrated) relative to a track of the image capturing portion. The image capturing portion may be operable to rotate three hundred and sixty degrees during image acquisition. The image capturing portion may rotate around a central point and/or axis, allowing image data of the patient to be acquired from multiple directions or in multiple planes.
(97) The O-arm® **106** with the gantry housing **124** has a central opening for positioning around an object to be imaged, a source of radiation that is rotatable around the interior of gantry housing **124**, which may be adapted to project radiation from a plurality of different projection angles. A detector system is adapted to detect the radiation at each projection angle to acquire object images from multiple projection planes in a quasi-simultaneous manner. The gantry may be attached to a

support structure O-arm® support structure, such as a wheeled mobile cart with wheels, in a cantilevered fashion. A positioning unit translates and/or tilts the gantry to a planned position and orientation, preferably under control of a computerized motion control system. The gantry may include a source and detector disposed opposite one another on the gantry. The source and detector may be secured to a motorized rotor, which may rotate the source and detector around the interior of the gantry in coordination with one another. The source may be pulsed at multiple positions and orientations over a partial and/or full three hundred and sixty degree rotation for multi-planar imaging of a targeted object located inside the gantry. The gantry may further comprise a rail and bearing system for guiding the rotor as it rotates, which may carry the source and detector. Both and/or either O-arm® **106** and C-arm **104** may be used as automated imaging system to scan a patient and send information to the surgical system **2**.
(98) Images captured by the automated imaging system can be displayed a display device of the surgical planning computer **910**, the surgical robot **800**, and/or another component of the surgical system **2**.
(99) Various embodiments of passive end effectors that are configured for use with a surgical system are now described in the context of FIGS. **12-19**.
(100) As will be explained in further detail below, the various passive end effectors illustrated in FIGS. **12-19** each include a base, a first planer mechanism, and a second planner mechanism. The base is configured to attach to an end effector coupler (e.g., end effector coupler **22** in FIGS. **4** and **5**) of a robot arm (e.g., robot arm **18** in FIGS. **1** and **2**) positioned by a surgical robot. Various clamping mechanisms may be used to firmly attach the base to the end effector coupler, removing backlash and ensuring suitable stiffness. Irreversible clamping mechanisms which may be used to attach the base to the end effector coupler can include but are not limited to toggle joint mechanisms or irreversible locking screw(s). A user may use an additional tool, such as but not limited to, a screwdriver, torque wrench, or driver to activate or tighten the clamping mechanism. The first mechanism extends between a rotatable connection to the base of a two and a rotatable connection to a tool attachment mechanism. The second mechanism extends between a rotatable connection to the base and a rotatable connection to the tool attachment mechanism. The first and second mechanisms pivot about the rotatable connections. The rotatable connections may be pivot joints allowing 1 degree-of-freedom (DOF) motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions. When pivot joints are used the first and second mechanisms can be configured to constrain movement of the tool attachment mechanism to a range of movement within a working plane. The tool attachment mechanism is configured to connect to a surgical saw having a saw blade that is configured to oscillate for cutting. The first and second mechanisms may be configured, e.g., via pivot joints having 1 DOF motion, to constrain a cutting plane of the saw blade to be parallel to the working plane. The tool attachment mechanism may connect to the surgical saw or saw blade through various mechanisms that can include, but are not limited to, a screw, nut and bolt, clamp, latch, tie, press fit, or magnet. A DRA can be connected to the tool attachment mechanism or the surgical saw to enable tracking of a pose of the saw blade by the camera tracking system **6** (FIG. **3**).
(101) As explained above, a surgical system (e.g., surgical system **2** in FIGS. **1** and **2**) includes a surgical robot (e.g., surgical robot **4** in FIGS. **1** and **2**) and a tracking system (e.g., camera tracking system **6** in FIGS. **1** and **3**) that is configured to determine a pose of an anatomical structure that is to be cut by the saw blade and to determine a pose of the saw blade. The surgical robot includes a robot base, a robot arm that is rotatably connected to the robot base and configured to position the passive end effector. At least one motor is operatively connected to move the robot arm relative to the robot base. At least one controller is connected to the at least one motor and configured to perform operations that include determining a pose of a target plane based on a surgical plan defining where the anatomical structure is to be cut and based on the pose of the anatomical structure, where the surgical plan may be generated by the surgical planning computer **910** of FIG.

9 based on input from an operator, e.g., surgeon or other surgery personnel. The operations further include generating steering information based on comparison of the pose of the target plane and the pose of the surgical saw. The steering information indicates where the passive end effector needs to be moved to position the working plane of the passive end effector so the cutting plane of the saw blade is aligned with the target plane.

(102) In some further embodiments, the operations performed by the at least one controller further include controlling movement of the at least one motor based on the steering information to reposition the passive end effector so the cutting plane of the saw blade becomes aligned with the target plane and the saw blade becomes positioned a distance from the anatomical structure to be cut that is within the range of movement of the tool attachment mechanism of the passive end effector.

(103) The operations may include providing the steering information to a display device for display to guide operator movement of the passive end effector so the cutting plane of the saw blade becomes aligned with the target plane and so the saw blade becomes positioned a distance from the anatomical structure, which is to be cut, that is within the range of movement of the tool attachment mechanism of the passive end effector.

(104) As explained above, some surgical systems can include head-mounted display devices that can be worn by a surgeon, nurse practitioner, and/or other persons assisting with the surgical procedure. The surgical systems can display information that allows the wearer to position the passive end effector more accurately and/or to confirm that it has been positioned accurately with the saw blade aligned with the target plane for cutting a planned location on an anatomical structure. The operation to provide the steering information to the display device, may include configuring the steering information for display on a head-mounted display device having a see-through display screen that displays the steering information as an overlay on the anatomical structure that is to be cut to guide operator movement of the passive end effector so the cutting plane of the saw blade becomes aligned with the target plane and the saw blade becomes positioned the distance from the anatomical structure within the range of movement of the tool attachment mechanism of the passive end effector.

(105) The operation to configure the steering information for display on the head-mounted display device, may include generating a graphical representation of the target plane that is displayed as an overlay anchored to and aligned with the anatomical structure that is to be cut, and generating another graphical representation of the cutting plane of the saw blade that is displayed as an overlay anchored to and aligned with the saw blade. A wearer may thereby move the surgical saw to provide visually observed alignment between the graphically rendered target plane and the graphically rendered cutting plane.

(106) The operation to configure the steering information for display on the head-mounted display device, may include generating a graphical representation a depth of cut made by the saw blade into a graphical representation of the anatomical structure being cut. Thus, the wearer can use the graphical representation of depth of cut to better monitor how the saw blade is cutting through bone despite direct observation of the cutting being obstructed by tissue or other structure.

(107) The tracking system can be configured to determine the pose of the anatomical structure that is to be cut by the saw blade based on determining a pose of tracking markers, e.g., DRAs, that are attached to the anatomical structure, and can be configured to determine a pose of the surgical saw based on determining a pose of tracking markers connected to at least one of the surgical saw and the passive end effector. The tracking system can be configured to determine the pose of the surgical saw based on rotary position sensors which are configured to measure rotational positions of the first and second mechanisms during movement of the tool attachment mechanism within the working plane. As explained above, position sensors may be directly connected to at least one joint of the passive end effector structure, but may also be positioned in another location in the structure and remotely measure the joint position by interconnection of a timing belt, a wire, or any other

synchronous transmission interconnection. Additionally the pose of the saw blade can be determined based on the tracking markers attached to the structure base, position sensors in the passive structure and kinematic model of the structure.

(108) The various passive end effectors disclosed herein can be sterilizable or non-sterile (covered by a sterile drape) passive 3 DOF (Degree Of Freedom) mechanical structures allowing mechanical guidance of a surgical saw or saw blade, such as a sagittal saw, along two translations in a plane parallel to the saw blade (defining the cut plane), and one rotation perpendicular to this cut plane (instrument orientation). During the surgery, the surgical robot **4** moves the end effector coupler **22**, and the passive end effector and surgical saw attached there, automatically to a position close to a knee or other anatomical structure, so that all bone to be cut is within the workspace of the passive end effector. This position depends on the cut to be made and the surgery planning and implant construction. The passive end effector can have 3 DOF to guide sagittal saw or saw blade on the cutting plane providing two translation (X and Y directions) and a rotation (around Z axis) as shown in FIG. **12**.

(109) When the surgical robot **4** achieves a planned position, it holds the position (either on brakes or active motor control) and does not move during the particular bone cut. It is the passive end effector that allows movement of the saw blade of the surgical saw along the planned target plane. Such planar cuts are particularly useful for classical total knee arthroplasty where all bone cuts are planar. In partial knee arthroplasty there are special types of implants, called “on-lay” which can be used in conjunction with saw-prepared bone surfaces. The various passive end effectors have mechanical structure that can ensure precision of guidance during cuts, with higher precision than classical jigs, and provide sufficient range of workspace range to cut all the bone that is planned and while provide sufficient transverse stiffness (corresponding to locked DOF) despite possibly significant amount of vibrations originating from the surgical saw in addition to forces applied by the surgeon and bone reactionary forces.

(110) As the same time, it is preferable to measure the passive end effector position because it enables the surgical robot **4** to inform the surgeon how much bone has been removed (procedure advancement). One way to provide real-time information on bone removal is for the surgical robot **4** to measure where the saw blade passed in reference to the bone because the blade can pass only where the bone has been cut. To measure saw blade position a DRA can be mounted to the surgical saw and/or the passive end effector. This enables direct or indirect measurement of the saw position in 3D space. An alternative way to measure saw blade position is to integrate position (rotation or translation) sensors (e.g. encoders, resolvers) into position information of the passive end effector in order to calculate position of the saw blade using a mathematical model of a defined relationship between location of the passive end effector geometry and the tip of the saw blade.

(111) In one embodiment, a conventional sagittal saw mechanism can be used with the surgical system computer platform **900** with little or no changes. The potential changes would involve adapting an external shield to enable easy attachment of the surgical saw to the passive end effector but would not necessarily involve changes in the internal mechanics. The passive end effector may be configured to connect to a conventional sagittal saw provided by, for example, DeSoutter company. In addition, the saw blade may be directly attached to the passive end effector without the saw handpiece.

(112) To prevent the saw from unintentional passive end effector movement when the surgical robot **4** positions the passive end effector, e.g., to prevent the surgical saw from falling on the patient due to gravitational forces, the passive end effector can include a lock mechanism that moves between engaged and disengaged operations. While engaged, the lock mechanism prevents movement of the saw blade with respect to the robot end effector coupler, either directly by locking the degree of freedoms (DOFs) of the surgical saw, or indirectly by braking or locking specifics joints of the passive end effector. While disengaged, the first and second mechanisms of the passive end effector can be moved relative to the base without interference from the lock mechanism. The

lock mechanism may also be used when a surgeon holds the surgical saw and controls the surgical robot **4** movement by applying forces and torques to the surgical saw. The surgical robot **4**, using the load cell **64** of FIGS. **6** and **7** integrated in the distal end of the robot arm **22**, measures forces and torques that are applied and generates responsive forces and torques on the robot arm **22** so the surgeon can more easily move the passive end effector back and forth, left and right, apply rotations around various axes.

(113) A first embodiment of a passive end effector is shown in FIG. **12**. Referring to FIG. **12**, the passive end effector **1200** includes a base **1202** configured to attach to an end effector coupler (e.g., end effector coupler **22** in FIGS. **4** and **5**) of a robot arm (e.g., robot arm **18** in FIGS. **1** and **2**) positioned by a surgical robot. The passive end effector **1200** further includes first and second mechanisms that extend between rotatable connections to the base **1202** and rotatable connections to a tool attachment mechanism. The rotatable connections may be pivot joints allowing 1 degree-of-freedom (DOF) motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions. The first and second mechanisms form a parallel architecture that positions the surgical saw rotation axis in the cut plane.

(114) First and second link segments **1210***a* and **1220***a* form the first planer mechanism, and third and fourth link segments **1210***b* and **1220***b* form the second planner mechanism. The first link segment **1210***a* extends between a rotatable connection to a first location on the base **1202** and a rotatable connection to an end of the second link segment **1220***a*. The third link segment **1210***b* extends between a rotatable connection to a second location on the base **1202** and a rotatable connection to an end of the fourth link segment **1220***b*. The first and second locations on the base **1202** are spaced apart on opposite sides of a rotational axis of the base color to when rotated by the robot arm. The tool attachment mechanism is formed by a fifth link segment that extends between rotatable connections to distal ends of the second link segment **1220***a* and the fourth link segment **1220***b* relative to the base **1202**. The first and second mechanisms (first and second link segments **1210***a***-1220***a* and third and fourth link segments **1210***b***-1220***b*) pivot about their rotatable connections to constrain movement of the tool attachment mechanism **1230** to a range of movement within a working plane. The tool attachment mechanism **1230** is configured to connect to a surgical saw **1240** having a saw blade **1242** that is configured to oscillate for cutting. The first and second mechanisms (first and second link segments **1210***a***-1220***a* and third and fourth link segments **1210***b***-1220***b*) may be configured, e.g., via pivot joints having 1 DOF motion, to constrain a cutting plane of the saw blade **1242** to be parallel to the working plane. The tool attachment mechanism **1230** may connect to the surgical saw **1240** through various mechanisms that can include, but are not limited to, a screw, nut and bolt, clamp, latch, tie, press fit, or magnet. A DRA **52** can be connected to the tool attachment mechanism **1230** or the surgical saw **1240** to enable tracking of a pose of the saw blade **1242** by the camera tracking system **6** (FIG. **3**).

(115) The passive end effector **1200** provides passive guidance of the surgical saw **1240** to constrain the saw blade **1242** to a defined cutting plane and reduce its mobility to three degrees of freedom (DOF): two translations Tx and Ty in a plane parallel to the cutting plane of the saw blade **1242**; and one rotational Rz around an axis perpendicular to the cutting plane.

(116) In some embodiments, a tracking system is configured to determine the pose of the saw blade **1242** based on rotary position sensors connected to the rotational joints of at least some of the link segments of the passive end effector **1200**. The rotary position sensors are configured to measure rotational positions of the joined link segments during movement of the tool attachment mechanism within the working plane. For example, a rotary position sensor can be configured to measure rotation of the first link segment **1210***a* relative to the base **1202**, another rotary position sensor can be configured to measure rotation of the second link segment **1220***a* relative to the first link segment **1210***a*, and another rotary position sensor can be configured to measure rotation of the tool attachment mechanism **1230** relative to the second link segment **1220***a*. The surgical saw **1240** can connected to have a fixed orientation relative to the tool attachment mechanism **1230**. A serial

kinematic chain of the passive end effector **1200** connecting the saw blade **1242** and the robot arm **22**, having serialized link segments and pivoting joints, provides the required mobility to the surgical saw **1240**. The position of the tip of the saw blade **1242** in the plane defined by the passive kinematic chain can be fully determined by the joint angles, sensed through the rotary position sensors, and the structural geometry of the interconnected link segments. Therefore, by measuring the relative angle between each connected link segment, for example along one or more interconnected paths between the base **1202** and the surgical saw **1240**, the position of the tip of the saw blade **1242** in the cut space can be computed using the proposed forward kinematic model. When the position and orientation of robot arm **22** distal end position and orientation with respect to the bone is known, the position and orientation of the saw blade **1242** with respect to the bone can be computed and displayed as feedback to the surgeon. For exemplary implementations where the saw blade is directly attached to the passive end effector, the frequency of measurement provided by rotary position sensors may be at least two times higher than saw blade oscillation frequency in order to measure saw blade position even during oscillations.

(117) Example types of rotary position sensors that can be used with passive end effectors herein can include, but are not limited to: potentiometers; optical; capacitive; rotary variable differential transformer (RVDT); linear variable differential transformer (LVDT); Hall effect; and incoder.

(118) A potentiometer based sensor is a passive electronic component. Potentiometers work by varying the position of a sliding contact across a uniform resistance. In a potentiometer, the entire input voltage is applied across the whole length of the resistor, and the output voltage is the voltage drop between the fixed and sliding contact. To receive and absolute position, a calibration-position is needed. Potentiometers may have a measurement range smaller 360°.

(119) An optical encoder can include a rotating disk, a light source, and a photo detector (light sensor). The disk, which is mounted on the rotating shaft, has patterns of opaque and transparent sectors coded into the disk. As the disk rotates, these patterns interrupt the light emitted onto the photo detector, generating a digital or pulse signal output. Through signal encoding on the disk absolute and relative as well as multi-turn measurements are possible.

(120) A capacitive encoder detects changes in capacitance using a high-frequency reference signal. This is accomplished with the three main parts: a stationary transmitter, a rotor, and a stationary receiver. Capacitive encoders can also be provided in a two-part configuration, with a rotor and a combined transmitter/receiver. The rotor can be etched with a sinusoidal pattern, and as it rotates, this pattern modulates the high-frequency signal of the transmitter in a predictable way. The encoder can be multi-turn, but absolute measurement is difficult to realize. Calibration at startup is needed.

(121) RVDT and LVDT sensors operate where the core of the transformer is in null position, the output voltages of the two, primary and secondary, windings are equal in magnitude, however opposite in direction. The overall output of the null position is always zero. An angular displacement from the null position is inducing a total differential output voltage. Therefore, the total angular displacement is directly proportional to the linear differential output voltage. The differential output voltages increase with clockwise and decrease with anti-clockwise direction. This encoder works in absolute measurement and may not be multi-turn compatible. Calibration during assembly is needed.

(122) In a Hall effect sensors, a thin strip of metal has a current applied along it. In the presence of a magnetic field, the electrons in the metal strip are deflected toward one edge, producing a voltage gradient across the short side of the strip, i.e., perpendicular to the feed current. In its simplest form, the sensor operates as an analog transducer, directly returning a voltage. With a known magnetic field, its distance from the Hall plate can be determined. Using groups of sensors, the relative position of the magnet can be deduced. By combining multiple sensor elements with a patterned magnet-plate, position can be detected absolute and relative similar to optical encoders.

(123) An incoder sensor works in a similar way to rotary variable transformer sensor, brushless

resolvers or synchros. The stator receives DC power and produces a low power AC electromagnetic field between the stator and rotor. This field is modified by the rotor depending on its angle. The stator senses the resulting field and outputs the rotation angle as an analogue or digital signal. Unlike resolvers, incoders use laminar circuits rather than wound wire spools. This technology enables incoders compact form, low mass, low inertia and high accuracy without high precision installation. A signal (Z) to count one full rotation is transmitted. Multi-turn and absolute sensing is possible.

(124) A second embodiment of a passive end effector is shown in FIG. **13**. Referring to FIG. **13**, the passive end effector **1300** includes a base **1302** that is configured to attach to an end effector coupler (e.g., end effector coupler **22** in FIGS. **4** and **5**) of a robot arm (e.g., robot arm **18** in FIGS. **1** and **2**) positioned by a surgical robot. The passive end effector **1300** further includes first and second mechanisms that extend between rotatable connections to the base **1302** and rotatable connections to a tool attachment mechanism. The rotatable connections may be pivot joints allowing 1 DOF motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions. First and second link segments **1310***a* and **1320***a* form the first planer mechanism, and third and fourth link segments **1310***b* and **1320***b* form the second planner mechanism. The first link segment **1310***a* extends between a rotatable connection to a first location on the base **1302** and a rotatable connection to an end of the second link segment **1320***a*. The third link segment **1310***b* extends between a rotatable connection to a second location on the base **1302** and a rotatable connection to an end of the fourth link segment **1320***b*. The first and second locations on the base **1302** are spaced apart on opposite sides of a rotational axis of the base color to when rotated by the robot arm. Distal ends of the second link segment **1320***a* and the fourth link segment **1320***b* from the base **1302** are rotatably connected to each other and to the tool attachment mechanism **1330**. The first and second mechanisms (first and second link segments **1310***a*-**1320***a* and third and fourth link segments **1310***b*-**1320***b*) may be configured, e.g., via pivot joints having 1 DOF motion, to rotate about their rotatable connections to constrain movement of the tool attachment mechanism **1330** to a range of movement within a working plane. The tool attachment mechanism **1330** is configured to connect to a surgical saw **1240** having a saw blade **1242** that is configured to oscillate for cutting. The first and second mechanisms (first and second link segments **1310***a*-**1320***a* and third and fourth link segments **1310***b*-**1320***b*) constrain a cutting plane of the saw blade **1242** to be parallel to the working plane. The tool attachment mechanism **1330** may connect to the surgical saw **1240** through various mechanisms that can include, but are not limited to, a screw, nut and bolt, clamp, latch, tie, press fit, or magnet. A DRA can be connected to the tool attachment mechanism **1330** or the surgical saw **1240** to enable tracking of a pose of the saw blade **1242** by the camera tracking system **6** (FIG. **3**).

(125) A third embodiment of a passive end effector is shown in FIG. **14**. Referring to FIG. **14**, the passive end effector **1400** includes a base **1402** that is configured to attach to an end effector coupler (e.g., end effector coupler **22** in FIGS. **4** and **5**) of a robot arm (e.g., robot arm **18** in FIGS. **1** and **2**) positioned by a surgical robot. The base **1402** includes first and second elongated base segments **1404***a* and **1404***b* that extend from spaced apart locations on opposite sides of a rotational axis of the base **1402** when rotated by the robot arm. The first and second elongated base segments **1404***a* and **1404***b* extend in a direction away from the end effector coupler of the robot arm when attached to the passive end effector **1400**. The passive end effector **1400** further includes first and second mechanisms that extend between rotatable connections to the elongated base segments **1404***a* and **1404***b* and rotatable connections to a tool attachment mechanism. One or more of the rotatable connections disclosed for this embodiment may be pivot joints allowing 1 DOF motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions.

(126) The first mechanism includes a first link segment **1411***a*, the second link segment **1410***a*, a third link segment **1420***a*, and a fourth link segment **1430***a*. The first and second link segments **1411***a* and **1410***a* extend parallel to each other between rotatable connections to spaced apart

locations on the first elongated base segment **1404***a* and spaced apart locations on the third link segment **1420***a*. An end of the third link segment **1420***a* is rotatably connected to an end of the fourth link segment **1430***a*.
(127) The second mechanism includes a fifth link segment **1411***b*, a sixth link segment **1410***b*, and a seventh link segment **1420***b*. The fifth and sixth link segments **1411***b* and **1410***b* extend parallel to each other between rotatable connections to spaced apart locations on the second elongated base segment **1404***b* and spaced apart locations on the seventh link segment **1420***b*. The tool attachment mechanism includes an eighth link segment **1440** that extends between rotatable connections to distal ends of the fourth and seventh link segments **1430***a* and **1420***b* from the base **1402**. In a further embodiment, the eighth link segment **1440** of the tool attachment mechanism includes an attachment member **1442** that extends in a direction away from the base **1402** to a rotatable connector that is configured to connect to the surgical saw **1240**. The attachment member **1442** extends from a location on the eighth link segment **1440** that is closer to the fourth link segment **1430***a* than to the seventh link segment **1420***b*.
(128) The first and second mechanisms (set of link segments **1411***a*, **1410***a*, **1420***a*, **1430***a* and set of link segments **1411***b*, **1410***b*, **1420***b*) may be configured to pivot about their rotatable connections to constrain movement of the tool attachment mechanism **1440** to a range of movement within a working plane. The tool attachment mechanism **1440** is configured to connect to the surgical saw **1240** having the saw blade **1242** that is configured to oscillate for cutting. The first and second mechanisms may be configured, e.g., via pivot joints having 1 DOF motion, to constrain a cutting plane of the saw blade **1242** to be parallel to the working plane. The tool attachment mechanism **1440** may connect to the surgical saw **1240** through various mechanisms that can include, but are not limited to, a screw, nut and bolt, clamp, latch, tie, press fit, or magnet. A DRA can be connected to the tool attachment mechanism **1440**, such as to the attachment member **1442**, or the surgical saw **1240** to enable tracking of a pose of the saw blade **1242** by the camera tracking system **6** (FIG. **3**).
(129) The passive end effector **1400** of FIG. **14** has a parallel architecture enabling positioning of the surgical saw about a rotation axis in the cut plane. Synchronized and/or different motion of lateral parallelograms allow positioning of the surgical saw rotation axis in the cut plane.
(130) A fourth embodiment of a passive end effector is shown in FIG. **15**. The passive end effector **1500** includes a base **1502** that is configured to attach to an end effector coupler (e.g., end effector coupler **22** in FIGS. **4** and **5**) of a robot arm (e.g., robot arm **18** in FIGS. **1** and **2**) positioned by a surgical robot. The base **1502** may include first and second elongated base segments that extend from spaced apart locations on opposite sides of a rotational axis of the base **1502** when rotated by the robot arm. The first and second elongated base segments extend away from each other. The passive end effector **1500** further includes first and second mechanisms that extend between rotatable connections to the base **1502** and rotatable connections to a tool attachment mechanism. One or more of the rotatable connections disclosed for this embodiment may be pivot joints allowing 1 DOF motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions.
(131) The first mechanism includes a first link segment **1510***a*. The second mechanism includes a second segment **1510***b*. The tool attachment mechanism includes a third link segment **1520**, a fourth link segment **1530**, a fifth link segment **1540***a*, a sixth link segment **1540***b*, and a seventh link segment **1550**. The first and second link segments **1510***a* and **1510***b* extend between rotatable connections to first and second locations, respectively, on the base **1502**, e.g., to first and second elongated base segments extending away from the base **1502**, to rotatable connections at opposite ends of the third link segment **1520**. The first and second locations on the base **1502** are spaced apart on opposite sides of a rotational axis of the base when rotated by the robot arm. The fourth link segment **1530** extends from the third link segment **1520** in a direction towards the base **1502**. The fifth and sixth link segments **1540***a* and **1540***b* extend parallel to each other between rotatable

connections to spaced apart locations on the fourth link segment **1530** and spaced apart locations on the seventh link segment **1550**. The seventh link segment **1550** is configured to have a rotatable connector that is configured to connect to the surgical saw **1240**.

(132) The first through sixth link segments **1510***a-b*, **1520**, **1530**, and **1540***a-b* may be configured to pivot about their rotatable connections to constrain movement of the seventh link segment **1550** to a range of movement within a working plane. The seventh link segment **1550** is configured to connect to the surgical saw **1240** having the saw blade **1242** that is configured to oscillate for cutting. The first through sixth link segments **1510***a-b*, **1520**, **1530**, and **1540***a-b* may be configured to constrain a cutting plane of the saw blade **1242** to be parallel to the working plane, e.g., via pivot joints having 1 DOF motion. The seventh link segment **1550** may connect to the surgical saw **1240** through various mechanisms that can include, but are not limited to, a screw, nut and bolt, clamp, latch, tie, press fit, or magnet. A DRA can be connected to the seventh link segment **1550** or the surgical saw **1240** to enable tracking of a pose of the saw blade **1242** by the camera tracking system **6** (FIG. **3**).

(133) A fifth embodiment of a passive end effector is shown in FIG. **16**. The passive end effector **1600** includes a base **1602** that is configured to attach to an end effector coupler (e.g., end effector coupler **22** in FIGS. **4** and **5**) of a robot arm (e.g., robot arm **18** in FIGS. **1** and **2**) positioned by a surgical robot. The passive end effector **1600** further includes first and second mechanisms that extend between rotatable connections to the base **1502** and rotatable connections to a tool attachment mechanism. The first mechanism includes a first link segment **1610***a*. The second mechanism includes a second segment **1610***b*. The tool attachment mechanism includes a third link segment **1620**, a fourth link segment **1630**, a fifth link segment **1640***a*, a sixth link segment **1640***b*, and a seventh link segment **1650**. One or more of the rotatable connections disclosed for this embodiment may be pivot joints allowing 1 DOF motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions.

(134) The first and second link segments **1610***a* and **1610***b* extend between rotatable connections to first and second locations, respectively, on the base **1602** to rotatable connections at opposite ends of the third link segment **1620**. The first and second locations on the base **1602** are spaced apart on opposite sides of a rotational axis of the base **1602** when rotated by the robot arm. The fourth link segment **1630** extends from the third link segment **1620** in a direction away from the base **1602**. The fifth and sixth link segments **1640***a* and **1640***b* extend parallel to each other between rotatable connections to spaced apart locations on the fourth link segment **1630** and spaced apart locations on the seventh link segment **1650**. The seventh link segment **1650** is configured to have a rotatable connector that is configured to connect to the surgical saw **1240**.

(135) The first through sixth link segments **1610***a-b*, **1620**, **1630**, and **1640***a-b* may be configured to pivot about their rotatable connections to constrain movement of the seventh link segment **1650** to a range of movement within a working plane. The seventh link segment **1650** is configured to connect to the surgical saw **1240** having the saw blade **1242** that is configured to oscillate for cutting. The first through sixth link segments **1610***a-b*, **1620**, **1630**, and **1640***a-b* may be configured to pivot while constraining a cutting plane of the saw blade **1242** to be parallel to the working plane. The seventh link segment **1650** may connect to the surgical saw **1240** through various mechanisms that can include, but are not limited to, a screw, nut and bolt, clamp, latch, tie, press fit, or magnet. A DRA can be connected to the seventh link segment **1650** or the surgical saw **1240** to enable tracking of a pose of the saw blade **1242** by the camera tracking system **6** (FIG. **3**).

(136) The passive end effector **1600** provides two perpendicular translational movements for positioning the surgical saw rotation axis in the cutting plane, and where the two translations are implemented by parallelograms.

(137) A sixth embodiment of a passive end effector is shown in FIG. **17**. The passive end effector **1700** includes a base **1702** that is configured to attach to an end effector coupler (e.g., end effector coupler **22** in FIGS. **4** and **5**) of a robot arm (e.g., robot arm **18** in FIGS. **1** and **2**) positioned by a

surgical robot. The passive end effector **1700** further includes first and second mechanisms that extend between rotatable connections to the base **1702** and rotatable connections to a tool attachment mechanism. One or more of the rotatable connections disclosed for this embodiment may be pivot joints allowing 1 DOF motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions. The first and second mechanisms are connected to provide translation along the radius of a parallelogram. The first mechanism includes first and second link segments **1710** and **1720***b*. The first link segment **1710** extends between a rotatable connection to the base **1702** and a rotatable connection to an end of the second link segment **1720***b*. The second mechanism includes a third link segment **1720***a*. The tool attachment mechanism includes a fourth link segment **1730**. The second and third link segments **1720***b* and **1720***a* extend away from the base **1702** and parallel to each other between rotatable connections to spaced apart locations on the first link segment **1710** and spaced apart locations on the fourth link segment **1730**. The fourth link segment **1730** comprises an attachment member **1732** that extends in a direction away from the base to a rotatable connector that is configured to connect to the surgical saw **1240**. The attachment member **1732** extends from a location on the fourth link segment **1730** that is closer to the third link segment **1720***a* than to the second link segment **1720***b*.

(138) The first through third link segments **1710**, **1720***b*, **1720***a* may be configured to pivot about their rotatable connections to constrain movement of the fourth link segment **1730** to a range of movement within a working plane. The fourth link segment **1730** is configured to connect to the surgical saw **1240** having the saw blade **1242** that is configured to oscillate for cutting. The first through third link segments **1710**, **1720***b*, **1720***a* pivot and may be configured to constrain a cutting plane of the saw blade **1242** to be parallel to the working plane. The fourth link segment **1730**, e.g., the attachment member **1732** thereof, may connect to the surgical saw **1240** through various mechanisms that can include, but are not limited to, a screw, nut and bolt, clamp, latch, tie, press fit, or magnet. A DRA can be connected to the fourth link segment **1730**, e.g., to the attachment member **1732**, or the surgical saw **1240** to enable tracking of a pose of the saw blade **1242** by the camera tracking system **6** (FIG. **3**).

(139) A seventh embodiment of a passive end effector is shown in FIG. **18**. The passive end effector **1800** includes a base **1802** that is configured to attach to an end effector coupler (e.g., end effector coupler **22** in FIGS. **4** and **5**) of a robot arm (e.g., robot arm **18** in FIGS. **1** and **2**) positioned by a surgical robot. The passive end effector **1800** further includes first and second mechanisms that extend between rotatable connections to the base **1802** and rotatable connections to a tool attachment mechanism. One or more of the rotatable connections disclosed for this embodiment may be pivot joints allowing 1 DOF motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions. The first mechanism includes a first link segment **1810***a*. The second mechanism includes a second link segment **1810***b*. The tool attachment mechanism includes a third link segment **1820**. The first and second link segments **1810***a* and **1810***b* extend between rotatable connections to first and second locations, respectively, on the base **1802** to rotatable connections at opposite ends of the third link segment **1820**. The first and second locations on the base **1802** are spaced apart on opposite sides of a rotational axis of the base **1802** when rotated by the robot arm. The third link segment **1820** includes an attachment member **1822** that extends in a direction away from the base **1802** to a rotatable connector that is configured to connect to the surgical saw **1240**. The attachment member **1822** extends from a location on the third link segment **1820** that is closer to the first link segment **1810***a* than to the second link segment **1810***b*. One or more of the rotatable connections disclosed for this embodiment may be pivot joints allowing 1 DOF motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions.

(140) The first and second link segments **1810***a* and **1810***b* may be configured to pivot about their rotatable connections between the base **1802** and the third link segment **1820** to constrain movement of the attachment member **1822** to a range of movement within a working plane. In

some other embodiments one or more of the rotatable connections can be universal joints allowing 2 DOF motions or ball joints allowing 3 DOF motions such that the movement is not constrained to the working plane. The attachment member **1822** is configured to connect to the surgical saw **1240** having the saw blade **1242** that is configured to oscillate for cutting. The first and second link segments **1810***a* and **1810***b* pivot while constraining a cutting plane of the saw blade **1242** to be parallel to the working plane. The attachment member **1822** may connect to the surgical saw **1240** through various mechanisms that can include, but are not limited to, a screw, nut and bolt, clamp, latch, tie, press fit, or magnet. A DRA can be connected to the third link segment **1820**, e.g., to the attachment member **1822**, or the surgical saw **1240** to enable tracking of a pose of the saw blade **1242** by the camera tracking system **6** (FIG. **3**).

(141) An eighth embodiment of a passive end effector is shown in FIG. **19**. The passive end effector **1900** includes a base **1902** that is configured to attach to an end effector coupler (e.g., end effector coupler **22** in FIGS. **4** and **5**) of a robot arm (e.g., robot arm **18** in FIGS. **1** and **2**) positioned by a surgical robot. The passive end effector **1900** further includes a first link segment **1910** and a second link segment **1920**. The first link segment **1910** extends between a rotatable connection to the base **1902** and a rotatable connection to one end of the second link segment **1920**. Another end of the second link segment **1920** is rotatably connected to a tool attachment mechanism. The rotational axes q**1**, q**2** and q**3** are parallel to each other so as to provide a planar cutting plane for the blade **1242**. Thus, the 3 DOF motion of the saw **1240** includes x-direction Tx, y-direction Ty and rotational direction about z-axis Rz. One or more of the rotatable connections disclosed for this embodiment may be pivot joints allowing 1 DOF motion, universal joints allowing 2 DOF motions, or ball joints allowing 3 DOF motions.

(142) The tracking markers **52** attached to the end effector base **1902** and the saw **1240** along with the tracking markers on the bones (e.g., tibia and femur) can be used to precisely and continuously monitor the real-time location of the blade **1242** and blade tip relative to the patient bone being cut. Although not explicitly shown in other figures, the tracking markers can be attached to the saw **1240** and all end effectors **1902** in all embodiments to track the location of the blade relative to the patient bone being cut. Although not shown, alternatively or in addition to the tracking markers, encoders can be positioned in each of the link segments **1910** and **1920** to determine precisely where the saw blade tip is at all times.

(143) Example Surgical Procedure

(144) An example surgical procedure using the surgical robot **4** in an Operating Room (OR) can include:

(145) Optional step: surgery is pre-operatively planned based on medical images 1. The surgical robot **4** system is outside the Operating Room (OR). The nurse brings the system to the OR when patient is being prepared for the surgery. 2. The nurse powers on the robot and deploys the robot arm. Nurse verifies precision of robotic and tracking systems. 3. In the case of a sterilized passive end effector, the scrub nurse puts a sterile drape on the robot arm and mounts the passive end effector with the sagittal saw on the robot arm. The scrub nurse locks the passive end effector with a lock mechanism. Scrub nurse attached DRAs to passive structure through the drape (if necessary). For a non-sterilized passive end effector, the drape is placed after attachment of the passive end effector on the robot arm, the DRAs are attached to the passive end effector with the drape intervening therebetween, and a sterile saw or saw blade is attached to the passive end effector with the drape intervening therebetween. The lock mechanism is engaged, in order to fix position of the saw blade with respect to the end effector coupler. 4. The surgeon attaches navigation markers to the patient's bone(s), e.g., tibia and femur. The bones are registered with the camera tracking system **6** using, e.g., Horn point-to-point algorithm, surface matching or other algorithms. A soft-tissue balance assessment may be performed, whereby the system allows surgeon to assess balance of soft tissue in the operating room, e.g., by tracking relative movement of femur and tibia when surgeon applies forces in different directions (e.g. varus/valgus stress).

Soft-tissue balance information can be used to alter surgical plan (e.g. move implant parts, change implant type etc.). 5. When surgeon is ready to cut the bone, the scrub nurse brings the surgical robot **4** to the operating table close to the knee to be operated and stabilizes the surgical robot **4** on the floor. The system may operate to guide nurse in finding robot **4** position so that all cut planes are in robot and passive structure workspace. 6. The surgeon selects on the screen of the surgical robot **4** the different parameters according to the planning of the surgery to do the first cut (bone to be cut, cutting plan desired, etc.). 7. The surgical robot **4** automatically moves the robot arm **22** to reposition the passive end effector so the cutting plane of the saw blade becomes aligned with the target plane and the saw blade becomes positioned a distance from the anatomical structure to be cut that is within the range of movement of the tool attachment mechanism of the passive end effector. 8. The surgeon unlocks the passive end effector. 9. The surgeon performs the cut constrained to the cutting plane provide by the passive end effector. The surgical robot **4** may provide real-time display of the tracked location of the saw blade relative to bone so the surgeon can monitor progress of bone removal. In one way, the tracking subsystem processes in real-time the location of the saw relative to the bone based on camera images and various tracking markers attached to the saw, robot arm, end effector, femur and tibia. The surgeon can then lock the passive end effector using the lock mechanism upon completion of the cut. 10. The surgeon selects on the screen the next cut to be executed and proceeds as before. 11. The surgeon may perform a trial implant placement and intermediate soft-tissue balance assessment and based thereon may change the implant plan and associated cuts. 12. Following completion of all cuts, the nurse removes the surgical robot **4** from the operating table and unmounts the passive end effector from the robot arm. 13. The surgeon places the implants and finishes the surgery.

(146) In step 9 above, a physician may have a difficult time to visually confirm the progress of the cut due to tissue and ligaments around the bone and debris being created from the cut, and other surgical instruments near the bone. Even if the visual confirmation may be acceptable, there are areas of the bone the physician cannot see such as the posterior portion of the bone being cut.

(147) Advantageously, one robotic system embodiment of the present invention provides a way for the physician to confirm the progress of the bone being cut in multiple dimensions. The camera tracking system **6** along with the tracking markers attached to the end effector base (**1100**, **1202**, **1302**, **1402**, **1502**, **1602**, **1702**, **1802**, **1902**), robotic arm **20** and saw (**1140**, **1240**) allows the tracking subsystem **830** and the computer subsystem **820** to calculate in real-time the precise position of the saw blade relative to bone so the surgeon can monitor progress of bone removal.

(148) FIG. **20** is a screenshot of a display showing the progress of bone cuts during a surgical procedure. FIG. **20** shows the subsystems **830** and **820** displaying three images: lateral, A-P and top views. In each image, the real-time location of the saw blade **1242** relative to the bone (e.g., tibia **2000**) is displayed on the display **34**. The lateral and top views can be especially useful for the physician as they display the saw blade position which cannot be easily seen. At the top portion of the display, the computer subsystem **830**, **820** displays the number of cutting programs and what program it is currently running. For example as the screenshot shows, the physician may have programmed **6** planar cuts and the current cutting program is the first one. Also, because the subsystems **830** and **820** can track with tracking markers where the blade may have travelled, it can determine how much of the bone cutting (area that was cut) for a particular cutting program has been completed and the percentage of progress is displayed in the display **34**. The bone image itself is preferably derived from actual images of the patient's body for a more accurate representation. The bone image is augmented by the subsystem **820** with a contour line that shows the cortical bone **2004** and spongy bone **2002**. This can be important for a physician as the amount of resistance to cutting varies greatly between the two types of bones.

(149) If an augmented reality (AR) head-mounted display is used, the computer subsystem **820** can generate the same contour line showing the cortical and spongy bones and superimpose it over the actual leg continuously as the physician moves his/her head. The area that has been already cut can

be overlaid over the actual bone in a dark shade. Moreover, the implant to be inserted over the cut area can also be overlaid on the bone to show the physician that the cutting is being done correctly along the plane of the implant. This is all possible because the subsystems **830** and **820** can track the position of the blade and history of its movement relative to the bone with the tracking markers and the camera subsystem.

(150) Referring now to FIGS. **21-32**, an exemplary embodiment of a direct blade guidance system in the context of orthopedic surgery is described. As shown in FIGS. **21**, **22**, **29**, and **30** a direct blade guidance system **2100** may include a robotic system holding an end effector arm (EEA) **2102**. EEA **2102** may include a base configured to attach to an end effector coupler of a robot arm. Other exemplary embodiments of an end effector arm consistent with the principles of this disclosure are described with regard to FIGS. **12-19**. In order to achieve planar cuts, a saw blade **2104** should be guided in the plane in which it vibrates. Saw blade high vibration frequency (e.g. 200-300 Hz) makes it difficult to realize mechanical guidance. EEA **2102** may include several joints and linkages **2106**, **2108** that may be used to realize movement on the plane of a tip **2110** of EEA **2102**. Similar to embodiments of FIGS. **12-18**, tip **2110** of EEA **2102** may have three (3) degrees of freedom: movement in two directions on the plane and rotation around the axis normal to that plane. EEA **2102** may allow planar cuts and approach the target bone from all angles. System **2100** may also include a handpiece **2112** and a blade adaptor **2114** each connected to EEA **2102** and blade **2104**.

(151) The concept of directly guiding saw blade **2104** includes aligning the rotation axis of a distal end of linkage **2110** (distal rotational joint **2116** of EEA **2102**) with a blade vibration axis. Sagittal saws are in the majority of implementation mechanisms which generate small rotational movement of the saw blade (vibration/oscillation) around an axis close to the saw blade attachment. By aligning the rotation axis with the distal rotational joint **2116** of EEA **2102** (at a distal end of linkage **2108**), joint **2116** may be configured to enable general saw handpiece rotation and enable saw blade vibration.

(152) Saw blade **2104** may be configured to be linked to the distal joint rotation axis of EEA **2102** by blade adaptor **2114**. Blade adaptor **2114** may be configured to tighten saw blade **2104**. By adapting the blade adaptor **2114**, different sagittal saws can be integrated into system **2100**. Exemplary configurations for blade adaptor **2114** consistent with the present disclosure are discussed below. An exemplary blade adaptor is illustrated in FIG. **31**.

(153) Permanent Fixation

(154) In a permanent fixation configuration, blade **2104** may be permanently clamped to blade adaptor **2114**. A user (e.g., a scrub nurse) can assemble blade **2104** to blade adaptor **2114** while a surgeon is preparing a surgical field. Exemplary permanent fixation configurations may include the following. Blade **2104** may be firmly clamped to blade adaptor **2114** using a component such as screws or spring loaded latches configured to provide a clamping force against blade **2104**. Blade **2104** may include an interface, such as through holes, that are configured to fix blade **2104** to blade adapter **2114**. Blade **2104** and blade adapter **2114** may be manufactured as a single reusable device that is then provided on EEA **2102**. Example implementation involves a metal blade attached to a PEEK or Stainless steel blade adapter. Assembly of the blade to the blade adapter is performed by the scrub nurse. Blade **2104** and blade adapter may also be manufactured as a disposable (single use) device delivered sterile. Example implementation involves a metal blade which is overmolded using plastic injection molding technology.

(155) FIG. **22** illustrates system **2200**, which is an exemplary embodiment consistent with the principles of the present disclosure. In this embodiment, the blade adaptor is permanently affixed to the saw blade in a base that can rotate around the saw blade rotation axis and a clamping component that clamps the blade to the blade adaptor base by means of screws. FIG. **23** illustrates a blade **2204**, a blade adaptor **2214**, a distal rotational joint **2216**, and a clamp **2218** consistent with this configuration. These components can be the same or similar to components previously

described with regard to FIG. **21**.
(156) Detachable Fixation
(157) In a detachable fixation configuration, the blade can be quickly attached to and detached from the blade adapter in the field. The blade adaptor integrates a clamping mechanism that might be either active (normally closed clamping mechanism that is open by electrical signal) or passive (quick-coupling and/or quick-release mechanism).
(158) Moment of Inertia
(159) Blade adapter **2114** may significantly increase the moment of inertia of the coupled “blade/blade adapter” rotating around blade rotation axis. As soon as blade adapter **2114** vibrates together with saw blade **2104**, the unbalanced inertia generates dynamic forces and torques that are exported to the mechanical structure as well as a surgeon's hand. To optimize the implementation, the inertia of the vibrating elements around the vibration axis may be minimized. That means the lighter and closer the mass of the vibrating elements to the vibration axis, the better. The moment of inertia of the saw blade elements about the vibration axis for the handpiece guidance concept is:
(160) $I_h - I_d + MD^2$
(161) With D the distance between the blade vibration axis and the distal joint rotation axis, M the weight of the blade (weight of the saw blade elements) and I.sub.d the moment of inertia of the saw blade elements about the blade vibration axis. The moment of inertia is then reduced by MD.sup.2 with the direct blade guidance concept, which is the minimum as the distance between the vibration axis and the blade vibration axis is 0 (the axes are combined).
(162) FIG. **24** illustrates a system using a standard jig **2402**. Under the principles discussed herein, direct blade guidance may provide more blade effective cutting length than those provided by standard jigs, such as jig **2402** illustrated in FIG. **24**. L.sub.G represents the guiding length (length on which the saw blade is guided) and L.sub.E represents the blade effective length. As the guiding length can be made shorter with the direct blade guidance concept, the blade effective length is consequently longer. This is may be more comfortable for a surgeon and the surgeon may cut through more bone.
(163) The precision of the cut is given by the rigidity of the different elements composing the robotic system from the floor to the tip of the saw blade, including the robot system, EEA **2102**, a handpiece sagittal saw, a saw blade oscillation mechanism (such as blade adaptor **2114** or **2214**), and saw blade **2104**, **2204**. With the direct blade guidance, the backlash and imprecisions of the handpiece may be significantly minimized or prevented altogether because the saw blade is directly tightened with the blade adaptor.
(164) Moreover, the direct blade guidance concept may be easier to integrate with various existing sagittal saws. There may be no need to build a component which is specific to the shape of the sagittal saw (manufacturer specific) because the blade adaptor is an element which only needs to tighten the saw blade whose shape may be simple. In addition, the blade adaptor is easy to sterilize. It is a small, light and relatively simple mechanical element which can be made in such a way that it does not have any narrow spaces and can be easily disassembled. As previously mentioned, blade including blade adaptor might be delivered as single reusable device.
(165) Tracking
(166) FIG. **25** illustrates a direct blade guidance system **2100** that may include a navigation marker or array of markers **2500** attached to handpiece **2112** of a sagittal saw. Marker **2500** allows the sagittal saw to be tracked by the camera, such as camera tracking system **6**. Optical tracking enables measurement of the saw position but may it may be difficult to measure a direct saw blade position during cutting (high frequency of blade vibration). Moreover, the tracking would directly measure the deflection of the saw handpiece **2112** with respect blade **2104**, due to the poor stiffness of the blade/saw handpiece connection.
(167) To measure a direct saw blade position, FIG. **26** illustrates system **2100** with encoders **2600**, **2602**, and **2604** integrated to the joints of EEA **2102**. Encoder measurement can be usually made at

higher frequency and rotational precision than using optical tracking. For example, the measurement frequency can be at least two times higher than saw blade oscillation. With a blade position signal, additional useful information may be extracted, such as saw vibration frequency, on/off state, saw blade blocking in bone (for example by interpreting the signal and its derivatives) and others.

(168) In exemplary embodiments consistent with the principles of this disclosure, vibrations of the sagittal saw handpiece held by the surgeon may be reduced. By decreasing vibrations of the handpiece, tactile feedback could be improved and cutting efficiency could be increased. As previously described, the vibrations exported to surgeon's hand result from the exported dynamic forces and torques generated by the unbalanced inertia of couple blade/blade adaptor around blade rotation axis.

(169) One approach to reduce the vibrations may be achieved by filtering the vibrations. In FIG. **27**, a damping element **2700** may be connected between handpiece **2112** and the EEA **2102**. This might be implemented using pieces of rubber, pneumatic or hydraulic cylinder.

(170) Another approach to reduce the vibrations may be achieved by dynamically balancing the inertia of the coupled saw/saw adaptor around the saw rotation axis. This can be implemented by dynamically compensating the exported forces and torques generated by the vibrating saw blade using a compensation inertia that performs the exact opposite movement with the same dynamics. FIGS. **28** and **32** illustrates exemplary embodiments where compensation inertia can be coupled to the main inertia to be balanced using a mechanical inverter **2800**.

(171) Referring now to FIGS. **33-45**, consistent with the principles of the present disclosure, a navigated pin guide driver with a handle and an attached reference element may be used to obviate the need for using a placement instrument and navigating a placement instrument. For example, a surgeon may hold the handle and the navigation system may track the reference element. The navigation system provides real-time feedback on the placement of the pins and a visual representation of the cuts, or a visual representation of the cut block with respect to the patient anatomy. The surgeon may use the navigated pin handle free hand or in conjunction with the passive end effector as previously described herein.

(172) A proposed workflow consistent with the principles of the present embodiment is illustrated in FIG. **33** as method **3300**. At step **3302**, a patient may be registered to the navigation system. At step **3304**, a navigated pin guide driver system is aligned to a targeted area of a bone using navigation system feedback. At step **3306**, cut block pins are driven into bone. At step **3308**, cut blocks are attached to the bone over the cut block pins and at step **3310** cuts are performed through the cut blocks.

(173) Referring to FIG. **34**, an exemplary schematic of a navigated pin driver system **3400** is illustrated. Navigated pin guide driver system **3400** may include a handle **3402** for the user to hold, a reference element **3404**, a distal tip **3406** for docking onto cortical bone, and 2 parallel pin guides (or guide tubes) **3408** separated by a bridge **3410**. Bridge **3410** maintains spacing to ensure that pin spacing matches that of the corresponding cut block. While FIG. **34** is illustrated with a single instrument that has two parallel pin guides **3408**, system **3400** may be constructed with only one pin guide **3408**. In order to optimize the viewing angle and ergonomics for a user, bridge **3410** and reference element **3404** can be rotated about an axis **3500** as shown in FIG. **35**. Reference element may include markers configured to be tracked by a navigation system, for example, optical markers detectable by an infrared camera.

(174) When using system **3400**, a user may register the patient to a navigation system and make an initial incision to expose the knee joint. Once exposed the user can position the navigated pin driver system as indicated by the navigation system. The user may then insert cut block pins, which can be conventional style pins, through pin guide tubes **3408** as shown in FIG. **36**.

(175) An example of how the navigated pin guide driver system **3400** can be used clinically is as follows: (1) plan the implant placement; (2) register the patient to the navigation system; (3) expose

the knee; and (4) use the navigated pin driver system **3500** to insert pins **3800** for the distal femur cut block as shown in FIGS. **37-38**.

(176) After steps (1)-(4) noted above, at step (5) a distal cutting block **3900** may be attached over pins **3800** as shown in FIG. **39**. As shown in FIG. **40**, at step (6) a distal resection may be performed. Shown in FIG. **41**, at step (7), using system **3400**, pins are inserted or holes created for localizing a 4 in 1 cut block **4200** which is attached to the resected section at step (8) and shown in FIG. **42**.

(177) After the 4 in 1 cut block **42** is attached, at step (9), anterior femoral, femoral chamfer, posterior femoral, posterior chamfer resections are made using 4 in 1 cut block **4200**. At step (10), using navigated pin guide driver system **3400**, pins are placed for proximal tibial cut block **4500**. At step (11), proximal tibial cut block **4500** is put in place and a resection is performed. At step (12), the total knee arthroplasty may be completed by inserting the appropriate implants. Alternatively and without departing from the principles of this disclosure, resection may be performed on the tibia prior to resection of the femur.

(178) Consistent with the principles of the present disclosure, navigated pin driver system **3500** may include a conforming style gripper instead of a serrated distal tip. The gripper may use granular jamming and be activated by a vacuum already present in the operating room.

(179) In an alternative for the user to directly position the pin guide and to simplify navigated pin guide placement, software may assist the user as explained in the following workflow. (1) Showing, via software, the target position of one pin guide (e.g. distal tip) on the bone to the user, for example a computer screen, an augmented reality (AR) headset, and a laser pointer. (a) Showing only a position or entry point, i.e. correct instrument rotation is not required at this stage. (b) Distal tip teeth/sharp edges help user for finding and maintaining correct entry point. (c) Once the position of instrument distal tip is within predefined safety limit from the target one (e.g. 1 mm), indicating to the user via software (e.g. showing green color) and moving to the next step. (2) Showing, via software, a target rotation of the pin guide allowing the user to find a correct trajectory for the first pin. (3) User places first pin. (4) Showing, via software, a target rotation around the first pin to define the correct trajectory for the second pin. (5) User places second pin and retracts guide along the pins

FURTHER DEFINITIONS AND EMBODIMENTS

(180) In the above-description of various embodiments of present inventive concepts, it is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of present inventive concepts. Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which present inventive concepts belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense expressly so defined herein.

(181) When an element is referred to as being “connected”, “coupled”, “responsive”, or variants thereof to another element, it can be directly connected, coupled, or responsive to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected”, “directly coupled”, “directly responsive”, or variants thereof to another element, there are no intervening elements present. Like numbers refer to like elements throughout. Furthermore, “coupled”, “connected”, “responsive”, or variants thereof as used herein may include wirelessly coupled, connected, or responsive. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Well-known functions or constructions may not be described in detail for brevity and/or clarity. The term “and/or” includes any and all combinations of one or more of the associated listed items.

(182) It will be understood that although the terms first, second, third, etc. may be used herein to describe various elements/operations, these elements/operations should not be limited by these terms. These terms are only used to distinguish one element/operation from another element/operation. Thus, a first element/operation in some embodiments could be termed a second element/operation in other embodiments without departing from the teachings of present inventive concepts. The same reference numerals or the same reference designators denote the same or similar elements throughout the specification.
(183) As used herein, the terms “comprise”, “comprising”, “comprises”, “include”, “including”, “includes”, “have”, “has”, “having”, or variants thereof are open-ended, and include one or more stated features, integers, elements, steps, components or functions but does not preclude the presence or addition of one or more other features, integers, elements, steps, components, functions or groups thereof. Furthermore, as used herein, the common abbreviation “e.g.”, which derives from the Latin phrase “exempli gratia,” may be used to introduce or specify a general example or examples of a previously mentioned item, and is not intended to be limiting of such item. The common abbreviation “i.e.”, which derives from the Latin phrase “id est,” may be used to specify a particular item from a more general recitation.
(184) Example embodiments are described herein with reference to block diagrams and/or flowchart illustrations of computer-implemented methods, apparatus (systems and/or devices) and/or computer program products. It is understood that a block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by computer program instructions that are performed by one or more computer circuits. These computer program instructions may be provided to a processor circuit of a general purpose computer circuit, special purpose computer circuit, and/or other programmable data processing circuit to produce a machine, such that the instructions, which execute via the processor of the computer and/or other programmable data processing apparatus, transform and control transistors, values stored in memory locations, and other hardware components within such circuitry to implement the functions/acts specified in the block diagrams and/or flowchart block or blocks, and thereby create means (functionality) and/or structure for implementing the functions/acts specified in the block diagrams and/or flowchart block(s).
(185) These computer program instructions may also be stored in a tangible computer-readable medium that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable medium produce an article of manufacture including instructions which implement the functions/acts specified in the block diagrams and/or flowchart block or blocks. Accordingly, embodiments of present inventive concepts may be embodied in hardware and/or in software (including firmware, resident software, micro-code, etc.) that runs on a processor such as a digital signal processor, which may collectively be referred to as “circuitry,” “a module” or variants thereof.
(186) It should also be noted that in some alternate implementations, the functions/acts noted in the blocks may occur out of the order noted in the flowcharts. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved. Moreover, the functionality of a given block of the flowcharts and/or block diagrams may be separated into multiple blocks and/or the functionality of two or more blocks of the flowcharts and/or block diagrams may be at least partially integrated. Finally, other blocks may be added/inserted between the blocks that are illustrated, and/or blocks/operations may be omitted without departing from the scope of inventive concepts. Moreover, although some of the diagrams include arrows on communication paths to show a primary direction of communication, it is to be understood that communication may occur in the opposite direction to the depicted arrows.
(187) Many variations and modifications can be made to the embodiments without substantially departing from the principles of the present inventive concepts. All such variations and

modifications are intended to be included herein within the scope of present inventive concepts. Accordingly, the above disclosed subject matter is to be considered illustrative, and not restrictive, and the appended examples of embodiments are intended to cover all such modifications, enhancements, and other embodiments, which fall within the spirit and scope of present inventive concepts. Thus, to the maximum extent allowed by law, the scope of present inventive concepts are to be determined by the broadest permissible interpretation of the present disclosure including the following examples of embodiments and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

Claims

1. A method for performing a total knee arthroplasty surgery using a navigated pin guide driver system, the method comprising: registering a knee of a patient to a navigation system for tracking position of the knee; aligning the navigated pin guide driver system to a target area of the knee of the patient using the navigation system, the navigated pin guide driver system including: a handle; a first pin guide tube attached to the handle and having a through hole for receiving a first pin, the through hole extending along a longitudinal axis of the first pin guide tube; a reference array of markers arranged in a predetermined pattern and attached to the first pin guide tube, the reference array configured to be tracked by the navigation system; and a tubular distal tip coupled to a distal end of the first pin guide tube, the tubular distal tip having a through hole for receiving the first pin and configured to dock into cortical bone; a bridge having a first end attached to the first pin guide tube; and a second pin guide tube attached to other end of the bridge, and having a through hole for receiving a second pin and spaced from the first pin guide tube such that the first and second pins receive corresponding holes in a cut block for cutting a femur or tibia, wherein the bridge and the reference array are configured to rotate about the longitudinal axis of the first pin guide tube; driving the first and second pins into the knee through the first and second pin guide tubes while being guided by the navigation system; placing the cut block over the driven pins; and performing cuts to the target area using the placed cut block.
2. The method of claim 1, wherein the first pin guide tube is parallel to the second pin guide tube.
3. The method of claim 2, wherein the second pin guide extends upwardly from the bridge without extending downwardly from the bridge.
4. The method of claim 3, wherein the tubular distal tip extends downwardly from the bridge.
5. The method of claim 4, wherein the bridge is configured to maintain a spacing between the first pin guide tube and the second pin guide tube.
6. The method of claim 1, wherein the cut block is a 4 in 1 cut block.
7. The method of claim 1, wherein the cut block is a proximal tibial cut block.
8. The method of claim 1, wherein the reference array includes a plurality of optical tracking markers that are visible to the navigation system.
9. The method of claim 8, wherein the reference array includes four tracking markers visible to the navigation system.
10. The method of claim 1, wherein the handle extends laterally from the longitudinal axis of the first pin guide tube.
11. A method for performing a total knee arthroplasty surgery using a navigated pin guide driver system, the method comprising: registering a knee of a patient to a navigation system for tracking position of the knee; aligning the navigated pin guide driver system to a target area of the knee of the patient using the navigation system, the navigated pin guide driver system including: a handle; a first pin guide tube attached to the handle and having a through hole for receiving a first pin, the through hole extending along a longitudinal axis of the first pin guide tube; a reference array of markers arranged in a predetermined pattern and rigidly attached to the first pin guide tube, the reference array configured to be tracked by the navigation system; and a tubular distal tip coupled

to a distal end of the first pin guide tube, the tubular distal tip having a through hole for receiving the first pin and configured to dock into cortical bone; a bridge having a first end attached to the first pin guide tube; a second pin guide tube attached to a second end of the bridge and spaced from the first pin guide and having a through hole for receiving a second pin such that the first and second pins receive corresponding holes in a cut block for cutting a femur or tibia, wherein the bridge and the reference array are configured to rotate about the longitudinal axis of the first pin guide tube; driving the first and second pins into the knee through the first and second pin guide tubes while being guided by the navigation system; placing the cut block over the driven pins; and performing cuts to the target area using the placed cut block.
12. The method of claim 11, wherein the navigated pin guide driver system includes a bridge connected between first and second pin guide tubes, and the second pin guide tube extends upwardly from the bridge without extending downwardly from the bridge.
13. The method of claim 12, wherein the tubular distal tip extends downwardly from the bridge.
14. The method of claim 13, wherein the bridge is configured to maintain a spacing between the first pin guide tube and the second pin guide tube.
15. The method of claim 11, wherein the cut block is a 4 in 1 cut block.
16. The method of claim 11, wherein the cut block is a proximal tibial cut block.
17. The method of claim 11, wherein the reference array includes a plurality of optical tracking markers that are visible to the navigation system.
18. The method of claim 17, wherein the reference array includes four and only four tracking markers arranged in a predetermined pattern and visible to the navigation system.
19. The method of claim 11, wherein the handle extends laterally from the longitudinal axis of the first pin guide tube.